



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Restructuring / Deregulation

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Special thanks to Adjunct Associate Professor Gro Klæboe

Overview

- Restructuring-history
- Roles and responsibilities
- Power Exchange vs Power Pool
- Market clearing as an optimization Problem
- Energy only and capacity Markets

The focus is on deregulating the electricity sector

REGULATED



DEREGULATED

Prices are all determined by the regulatory/government bodies:

Vertically integrated structure

Cannot choose supplier

Prices are determined by “invisible hand” of the market

- energy prices
- transmission and distribution prices

Horizontal restructuring

Competition among a set of suppliers and retailers

Globally, from the early days until today

[Chicago Boys (ca. 1957) - Photo by Ernesto Fontaine]

1980s



- First ideas for liberalization of the electricity sector
- Introduction of electricity market concepts in Chile (influence of the “Chicago Boys”)

Globally, from the early days until today

1990



- UK (under the leadership of Margaret Thatcher) privatizes the electricity supply industry
- ... to be followed by other Commonwealth member countries, e.g. New Zealand and Australia

Globally, from the early days until today

1991



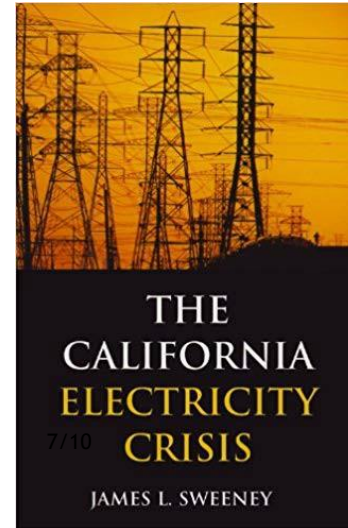
- Beginning of deregulation in Scandinavia...
- ... to be further detailed since zooming on Scandinavia in a few slides

Globally, from the early days until today

1996



2000-2001



- Deregulation in California... California electricity crisis!
- *In short:* a shortage of electricity supply, rise in prices (burnout prices), multiple black-outs, state of emergency, bankruptcies, investigation of Enron's role

Forslag til endringer i energiloven:
Tiltak for raskere
konsesjonsprosesser for kraft og
nett

Pressemelding | Dato: 07.02.2025

Lov om produksjon, omforming, overføring, omsetning, fordeling og bruk av energi m.m. (energiloven)

Dato	LOV-1990-06-29-50
Departement	Energidepartementet



Kort om loven

Sist endret 17.06.2024

Energiloven er en norsk lov som regulerer produksjon, omforming, overføring, omsetning, fordeling og bruk av energi, det vil si elektrisk kraft og fjernvarme. Kraftforsyningen er samfunnskritisk infrastruktur. Formålet med loven er å sikre at produksjon, omforming, overføring, omsetning, fordeling og bruk av energi foregår på en samfunnsmessig rasjonell måte. Her står samfunnsøkonomiske beregninger sterkt, men det skal også legges vekt på andre, ikke-kvantifiserbare hensyn som natur og miljø.

Bakgrunnen for loven var at man ønsket å rydde opp i og forenkle energiregelverket, som var spredt rundt i forskjellige lover, og et ønske om å få på plass regler for effektivisering av kraftmarkedet og en bedre organisering av energiforsyningen.

[Les mer i Store norske leksikon.](#)

<https://lovdata.no/dokument/NL/lov/1990-06-29-50?q=energiloven>

Norwegian power market before and after the Energy Act

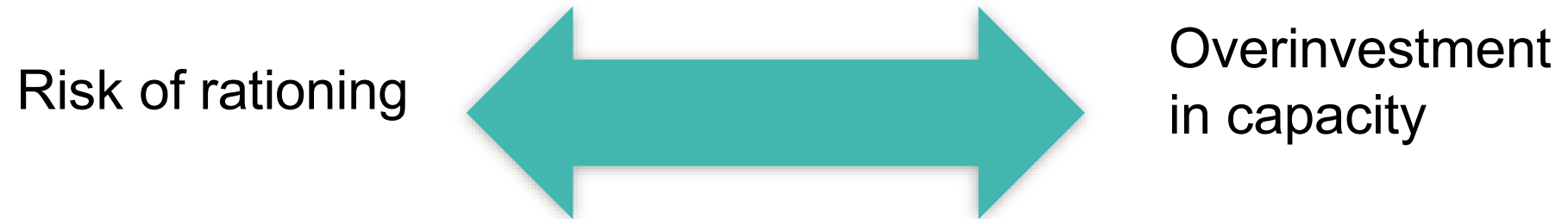
	Regulated	Deregulated
Electricity supplier	Assigned based on area (regional utility)	Free choice. Distribution grid provider assigned based on area
Utility ownership	Public	Mostly public
Pricing	Fixed, based on cost recovery	Variable from hour to hour, based on marginal cost
Market places	Wholesale market for excess supply	Most power is traded in the wholesale market
Investment	Motivated by energy/power sufficiency	Motivated by profits
Security of supply	Responsibility of each regional utility	Government has monitoring responsibility

Arguments for deregulation

- Avoid excessive investment
- Better efficiency – avoid regional overinvestment
- Incentivize cost reduction and more efficient system
- Equal prices for households and industry

Deregulation and security of supply

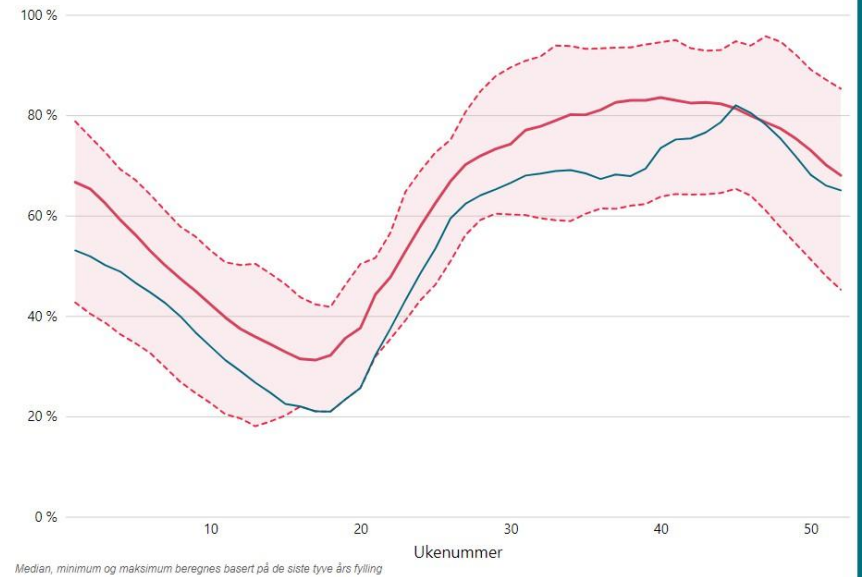
- In Norway, before deregulation, each regional utility was responsible for security of supply in that area
- Utilities built capacity for the driest/coldest years
- After deregulation very few measures are available to ensure security of supply



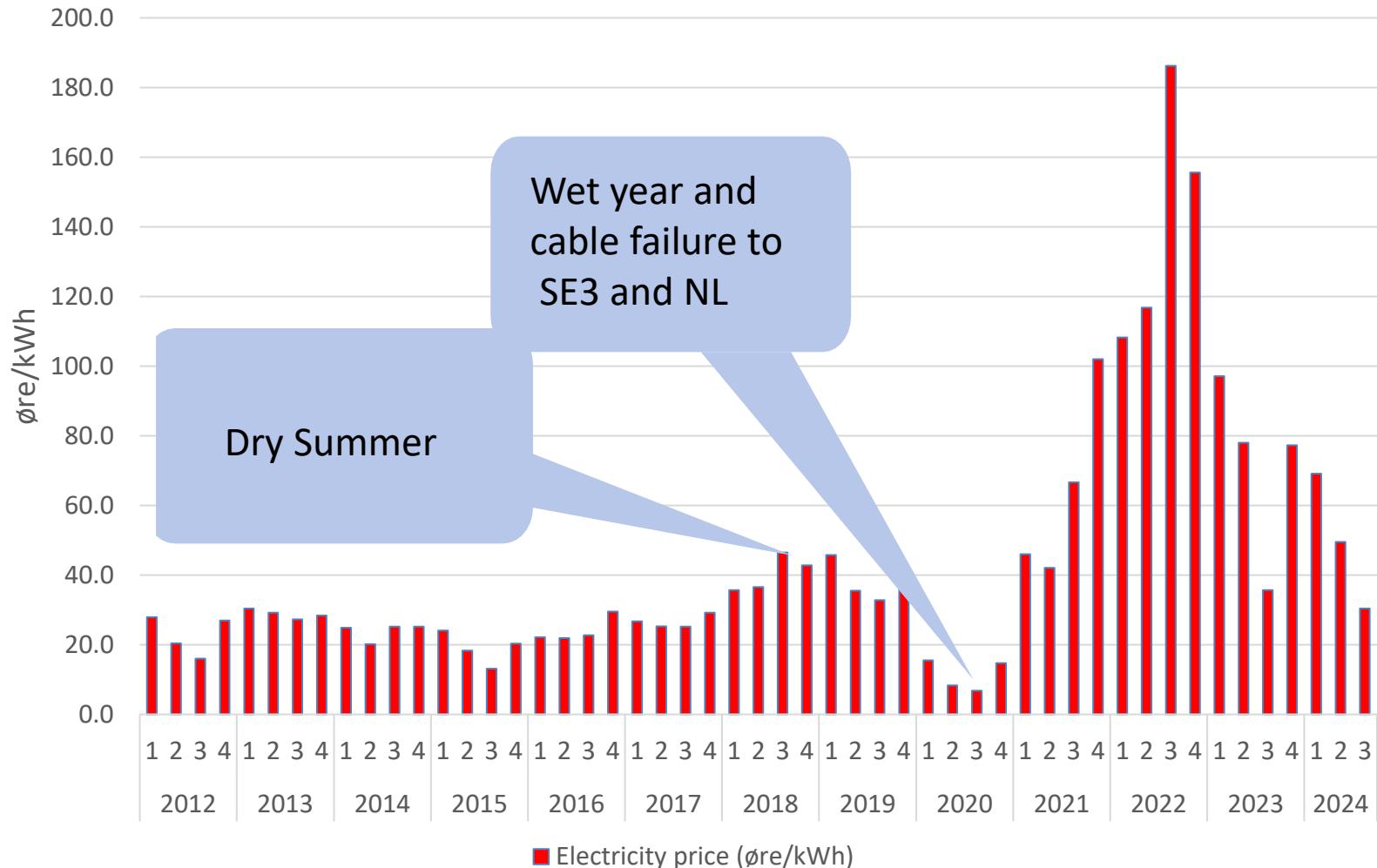
Spring/summer 2022

- Summer 2021: Very dry in Southern Norway, which continued into a dry winter and spring.
- Little import from Europe due to high gas prices.
- Producers in NO2, NO1 and NO5 was forced to report weekly to NVE how much storable water they used.

Fyllingsgrad Norge, 2022



Nordic wholesale power prices- Average quarterly

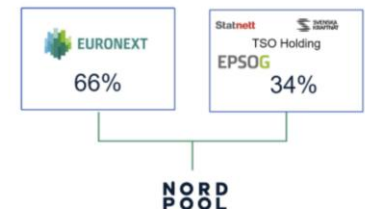


A history of Nord Pool

- 1991** Deregulation of the Norwegian electricity market
- 1996** Norwegian-Swedish exchange called **Nord Pool**
- 1998** Inclusion of Finland
Western Denmark joins the Nordic power exchange area
- 2000** Eastern Denmark's turn to join
- 2002** Nord Pool Spot established as a new and separate entity
- 2008** Nord Pool is acquired by NASDAQ
- 2009** Market coupling between Scandinavia and Germany
Negative prices accepted
- 2013** All Baltic countries have joined Nord Pool (Estonia-2010, Lithuania-2012, Latvia-2013)
- 2014** North-Western European power markets are coupled through the Price Coupling of Regions (PCR) project.
- 2015** Nord Pool Spot is appointed Nominated Electricity Market Operator (NEMO)
- 2016** Nord Pool Spot is rebranded to Nord Pool.
- 2017** Nord Pool introduced a new GB half hourly auction
- 2018** NODES - the marketplace of the future is established by Nord Pool and Agder Energi.
European Cross-Border Intraday (XBID) Solution
- 2019** Nord Pool launched a day-ahead power trading for central and western Europe (CWE)
- 2020** Euronext acquired a 66% stake in Nord Pool, Expand Intraday market to Poland
- 2021** Expanded day-ahead market into Poland
- 2025** Launched flow-based market coupling in Core Region



Nord Pool is owned by Euronext (66%) and TSO Holding (34%).



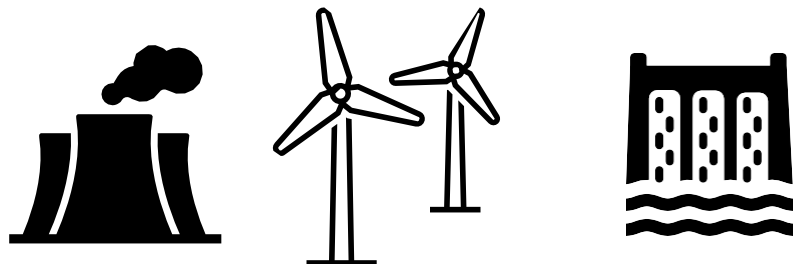
More on the history of Nord Pool (<https://www.nordpoolgroup.com/en/About-us/History/>)

Role and Responsibilities in Electricity Markets

Roles/responsibilities

- Power Exchange PX ([NordPool](#))
- Producers ([Statkraft](#), [E-CO](#), [Lyse Produksjon AS](#),)~225 companies
- Regulator ([NVE](#))
- Retailers ([Hafslund](#), [Tibber](#), [Aneo](#))
- End users ([Norsk Hydro](#))
- System Operator (ISO or TSO; [Statnett](#))
- Grid companies ([TENSIO](#), [Elvia](#), [Hålogaland Kraft Nett](#))
- Balance responsible parties (BRP)
- Traders and brokers
- Market surveillance, competition authorities ([Konkurransetilsynet](#))
- Imbalance Settlement Responsible

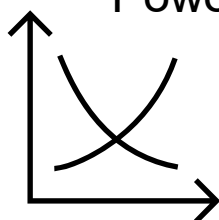
Roles in the Nordic power market



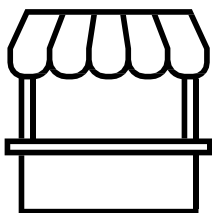
Generation companies



Balancing responsables



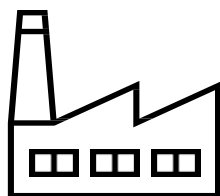
Power exchanges



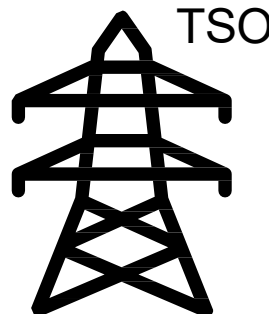
Retailers



End users



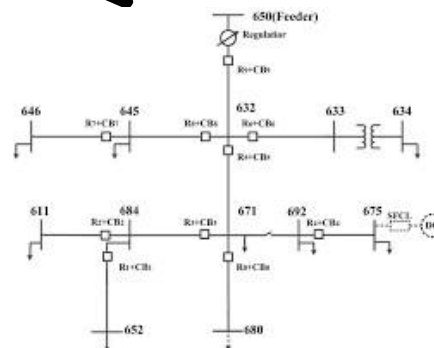
Statnett



TSO



Regulators

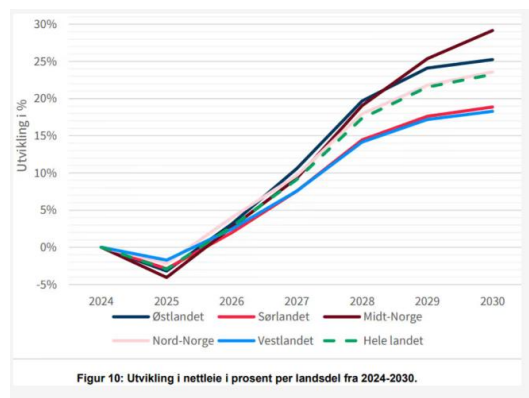


Grid companies

Traders and brokers

Regulator

- Government body, monitors and enforces laws and regulations
- Ensuring market efficiency and security of supply



Ny NVE-rapport: Nettleien vil skyte kraftig i været mot 2030

Hvor mye dyrere det blir avhenger av hvor du bor.



Torsdag 12. september 2024 kl. 08:44



Betaler du for mye for strøm? Sjekk ut VGs strømguide.

Snittet for hele landet blir trolig rett under 25 prosent.

Solenergi



RME åpner for at bedrifter og eneboliger skal kunne dele solkraft

Publisert 6. februar 2024 14:38

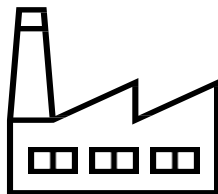


– Misforståelser om batterier

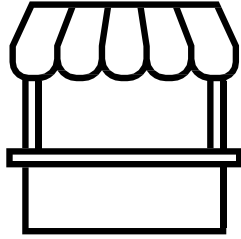
RME mener en artikkel i Europower om batterier gir et feil inntrykk. Det skriver RMEs Tiril Henriksen Norvoll i et debattinnlegg.

End users

- End users have the privilege of observing the price before consumption
- End users choose an electricity supplier (and price scheme) freely
- AMS meters enables billing hour by hour
- Capacity based grid tariffs discourage too high peak tariffs
- Grid supplier is assigned based on region



Retailers/suppliers



- 78 retailers currently operating in Norway
- Sells power to end users - either spot price, or various fixed price products
- Normally also balancing responsible for end users
- Margin business

talkmore
strøm



elekt

tibber



Balancing responsible parties

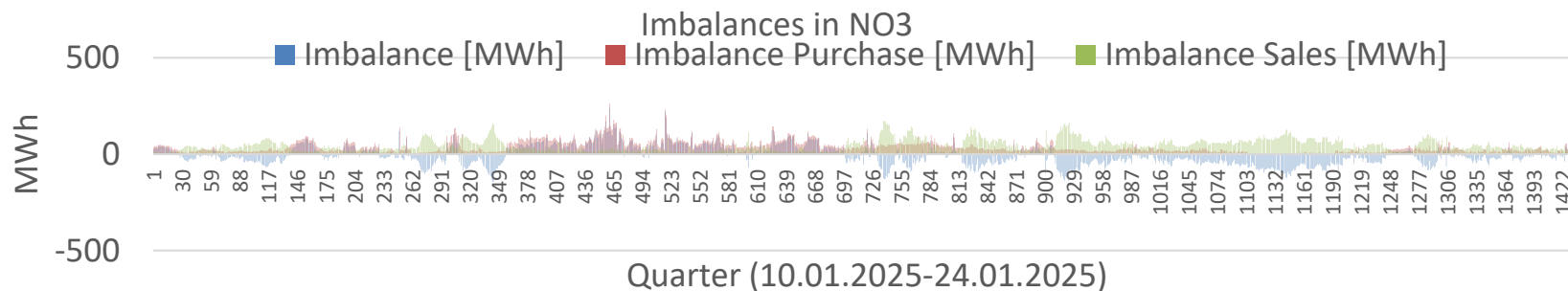
- Legal entity responsible for minimising the imbalance between obligations (from day-ahead market, bilateral) and actual consumption/production
- Normally retailers are balancing responsible for endusers and generation companies are their own balancing responsible.
- Tools: Trading in the intraday market, utilizing own flexibility
- Pays imbalance fee and imbalance cost
- With too much imbalance, the balancing responsible can lose operating license.

How much imbalance is accepted?

- Esett is the organisation for common settlement in the Nordic Countries
- Consult their handbook for questions about roles and imbalance

<https://www.esett.com/handbook/>

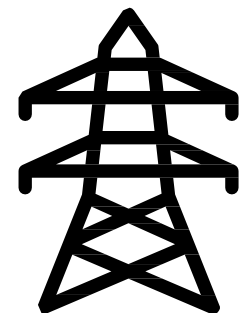
PU Type	Threshold	Country's threshold value		
		Finland	Norway	Sweden
Hydro	● $PP_1 - 1$. Threshold	4 %	2,5 %	2,5 %
	● $PP_2 - 2$. Threshold	6 %	6 %	6 %
Nuclear	● $PP_1 - 1$. Threshold	1 %	–	2,5 %
	● $PP_2 - 2$. Threshold	2 %	–	6 %
Other	● $PP_1 - 1$. Threshold	7 %	2,5 %	2,5 %
	● $PP_2 - 2$. Threshold	15 %	6 %	6 %
Solar	● $PP_1 - 1$. Threshold	18 %	15 %	6 %
	● $PP_2 - 2$. Threshold	25 %	25 %	12,5 %
Thermal Power	● $PP_1 - 1$. Threshold	3 %	2,5 %	2,5 %
	● $PP_2 - 2$. Threshold	6 %	6 %	6 %
Wind	● $PP_1 - 1$. Threshold	18 %	15 %	6 %
	● $PP_2 - 2$. Threshold	25 %	25 %	12,5 %



(Transmission) system operator

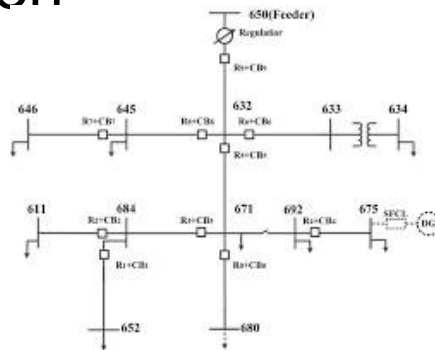
- Responsible for security of supply
- In the Nordics, the system operator is also responsible for the operation of the transmission network
- Responsibilities:
 - Keeping the grid frequency with limits
 - Defining transfer limits in the grid
 - Countertrading to relieve internal bottlenecks in the grid
 - Designing and operating balancing markets

Statnett



Grid companies

- There are approx. 100 distribution grid companies in Norway
- Responsibilities
 - Operation and maintenance of the grid
 - Connection of end users
 - Upgrades and investment planning
 - Metering and meter data handling for settlement
- Income comes from grid tariffs and is subject to public regulation



Traders & brokers

- Financial traders have a role in risk mitigating
- Power Purchase Agreements (PPA) are common in project financed investments in generation capacity

Tre krafttradere i Energi Danmark opparbeidet seg i fjor mellom 375 og 450 millioner norske kroner hver. I Norge holder Statkraft fortsatt hemmelig hvor mye deres tradere tjente.

28. mars 2023 5:01 OPPDATERT 28. mars 2023 5:01

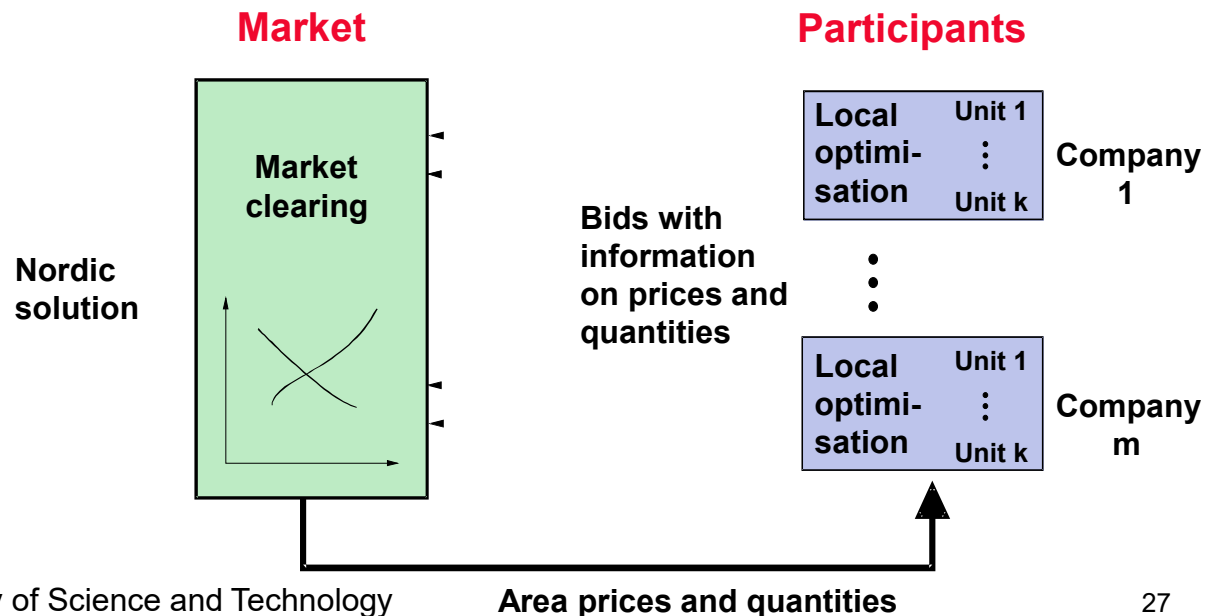
Av **Haakon Barstad** 🔔



Power Exchange vs Power Pool

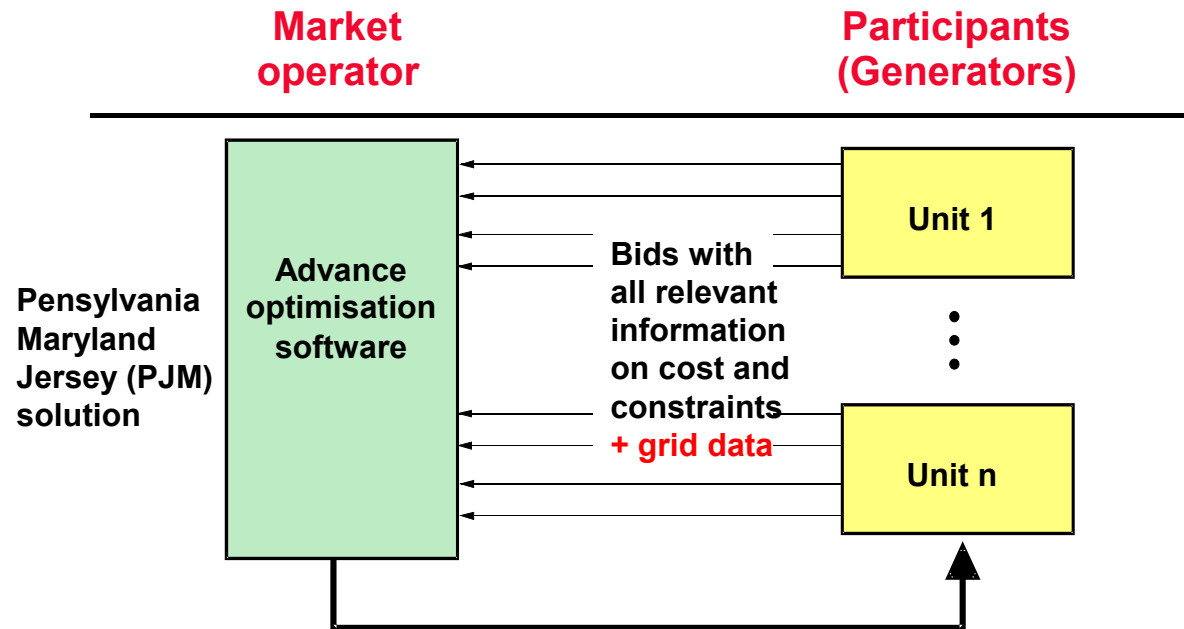
Power Exchange

- Decentralized Trading
 - Generators (Companies 1 ... m) each do their own “Local optimization” of power plants.
 - They submit bids with price and quantity information to the Market Clearing mechanism (the power exchange).
- Market Clearing
 - The power exchange aggregates all bids from buyers and sellers.
 - It matches supply and demand to determine a market clearing price and traded quantities for each bidding zone or area.
- Area Prices
 - Instead of detailed unit-by-unit dispatch by a central operator, the market produces area prices and quantities (one price for each region or zone).
 - Companies then decide how to run their portfolio of units to fulfill the obligations they accepted in the market.



Power Pool

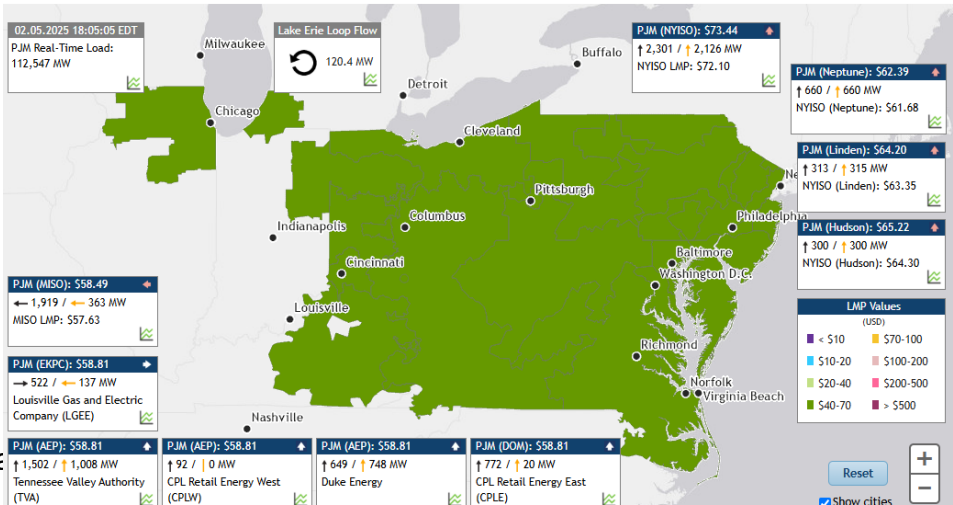
- Centralized Dispatch:
 - A central “Market Operator” runs advanced optimization software that collects detailed bids from each generating unit (Unit 1 ... Unit n).
 - These bids include cost data, technical constraints, and grid data (e.g., network limits).
- Schedules and Nodal Prices
 - Using this information, the market operator optimally dispatches all generators to meet demand reliably at the lowest system cost.
 - The outcome is a set of schedules (who produces how much and when) and nodal prices (electricity prices at specific points or “nodes” on the grid).



Power Pool vs. Power Exchange in Electricity Markets

Aspect	Power Pool	Power Exchange
Structure	Centrally coordinated, often mandatory	Decentralized trading platform, usually voluntary
Dispatch	Central entity dispatches generation to meet demand	Dispatch is not inherently centralized; trades inform schedules but system operator still manages reliability in real-time
Price Formation	Nodal pricing or locational marginal pricing (LMP)	Zonal pricing
Market Flexibility	Less flexible, more regulated, historically “all-in”	More flexible, participants can trade in different timeframes or venues
Examples	PJM	Nord Pool (Nordic), EEX (Central Europe), many modern wholesale electricity spot markets

- ✓ **Power Pool:** Centralized, mandatory participation with nodal pricing; dispatch is controlled by one operator.
- ✓ **Power Exchange:** Voluntary marketplace where buyers and sellers trade freely and discover prices through their bids



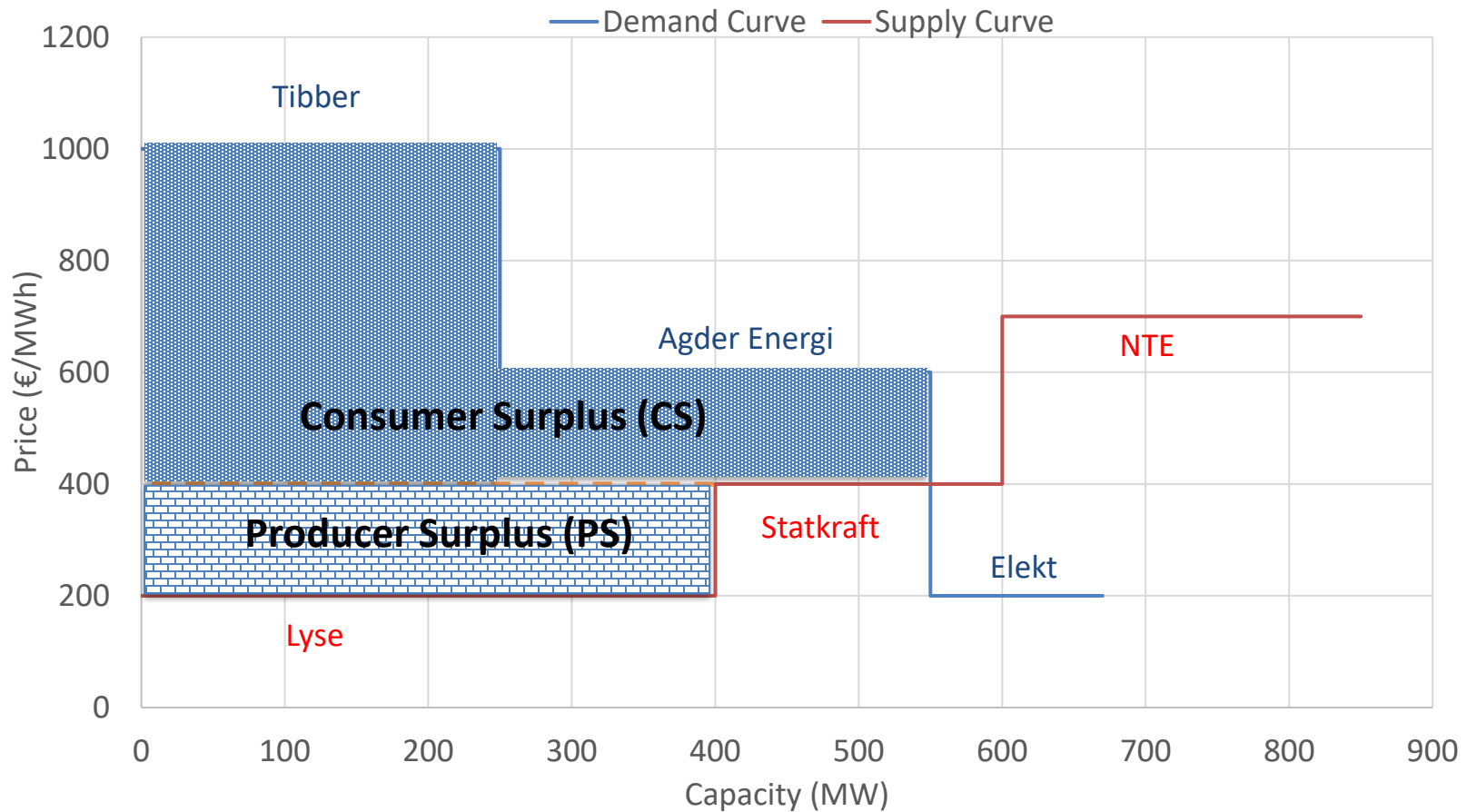
Market clearing as an optimization Problem

Optimization example – Market clearing

Name (Role)	Capacity [MWh]	Marginal cost [NOK/MWh]
Lyse (Producer)	$P_{g1}=400$	$C_1=200$
Statkraft (Producer)	$P_{g2}=200$	$C_2=400$
NTE (Producer)	$P_{g3}=250$	$C_3=700$
Tibber (Consumer)	$P_{d1}=350$	$U_1=1000$
Agder Energi (Consumer)	$P_{d2}=300$	$U_2=600$
Elekt (Consumer)	$P_{d3}=500$	$U_3=200$

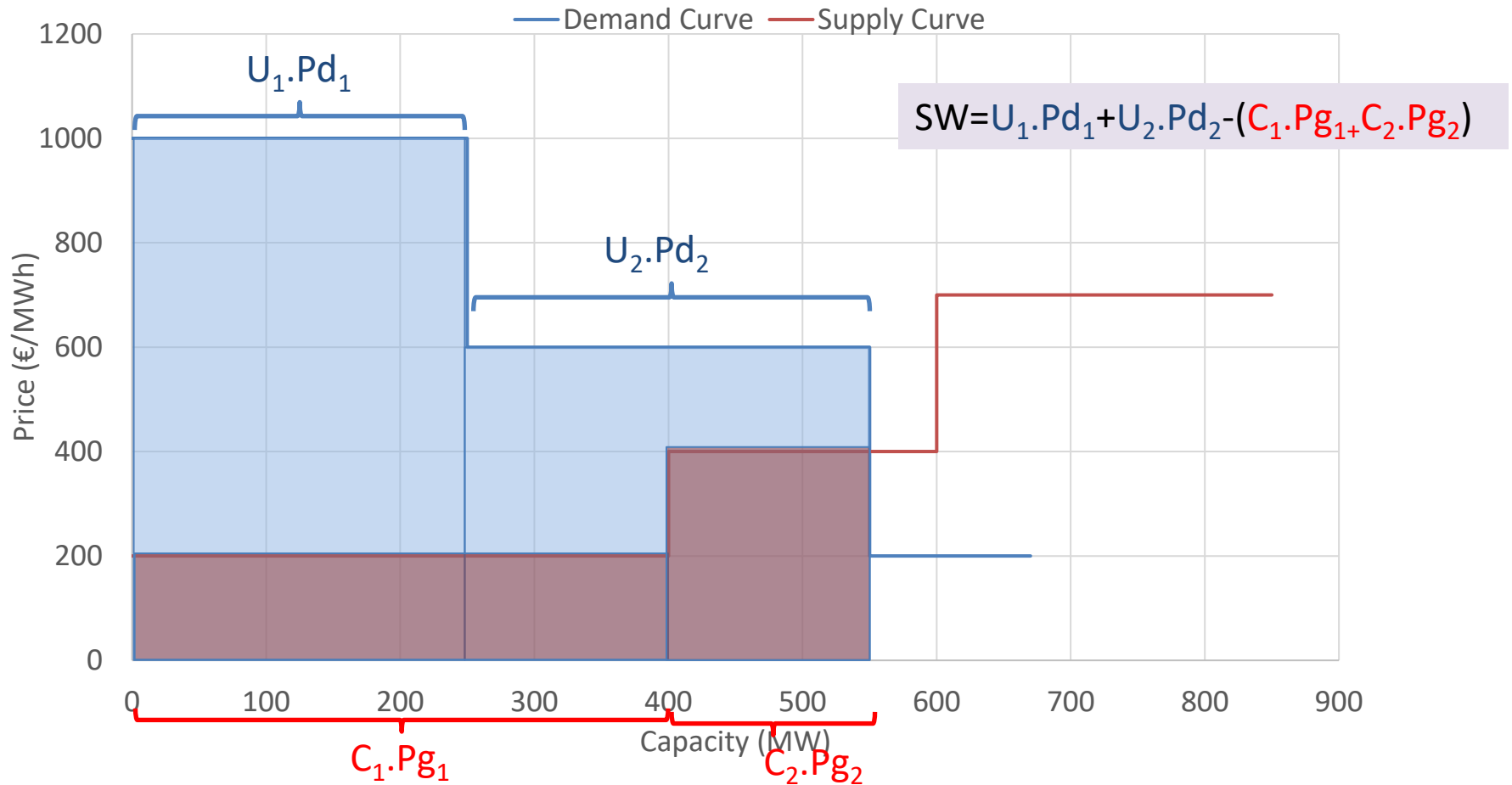
- 3 producers and 3 consumers offer/bid their electricity production/consumption for pool trading
- Each company has a defined capacity [MWh], and a marginal cost associated with the electricity [NOK/MWh]

Supply and Demand Curves



Social Welfare=PS +CS

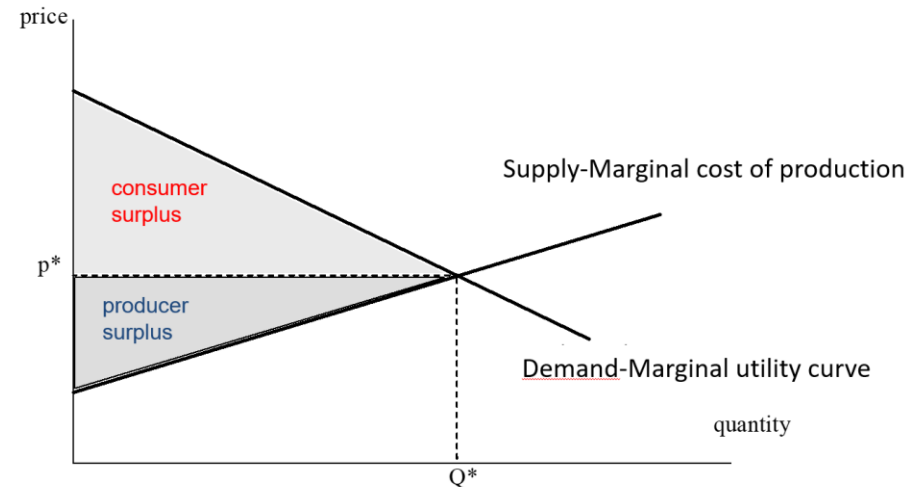
Supply and Demand Curves



In general, **Social welfare= Utility of demands - Cost of electricity production**

Market clearing – What do we aim to do?

- Defining decision variables:
 - What is it we want to find values for?
 1. p_g^G - Production quantity from producer p
 2. p_d^D - Consumption quantity from consumer c
- What is the objective function?
 - Maximize social welfare (CS+PS)
- What are constraints?
 1. Power balance i.e.,
Supply = Demand
 1. Supply limitation
 2. Demand limitation



Introducing notations

Inputs:

- All bids and offers in the market are formulated in terms of a *quantity* p and a *price* U_d and C_g for demand and supply, respectively.
- On the demand side (N_D demand bids):
 - set of bids: $P^D = \{p_d^D, d = 1, \dots, N_D\}$
 - maximum load of demands $p_d^D: \bar{P}_d^D$
 - bid prices for demand $p_d^D: U_d$
- On the supply side (N_G supply offers):
 - set of offers: $P^G = \{p_g^G, g = 1, \dots, N_G\}$
 - maximum capacity of generators $p_g^G: \bar{P}_g^G$
 - offer prices for generators $p_g^G: C_g$

Decision variables:

- *Consumption schedule*: $\mathbf{P}^D = [p_d^D, \dots, p_{N_D}^D]^T, 0 \leq p_d^D \leq \bar{P}_d^D$
- *Generation schedule*: $\mathbf{P}^G = [p_g^G, \dots, p_{N_G}^G]^T, 0 \leq p_g^G \leq \bar{P}_g^G$

Centralized social welfare optimization

- The social welfare maximization problem can be written as

$$\begin{aligned} \text{Max}_{p_g^G \geq 0, p_d^D \geq 0} \quad SW = & \underbrace{\sum_{d=1}^{N_D} U_d p_d^D}_{\text{Utility of demands}} - \underbrace{\sum_{g=1}^{N_G} C_g p_g^G}_{\text{Cost of electricity production}} \quad (1a) \end{aligned}$$

subject to:

$$\sum_{d=1}^{N_D} p_d^D - \sum_{g=1}^{N_G} p_g^G = 0 \quad (\lambda) \quad (1b) \rightarrow \text{Power balance}$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \quad (\gamma_g^G) \quad (1c) \rightarrow \text{Generation limits}$$

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \quad (\gamma_d^D) \quad (1d) \rightarrow \text{Consumption limits}$$

- λ and γ are sets of Lagrange multipliers associated with all equality and inequality constraints
- And equivalently as a minimization problem by minimizing the opposite objective function, i.e.

$$\begin{aligned} \text{Min}_{p_g^G \geq 0, p_d^D \geq 0} \quad SW = & \sum_{g=1}^{N_G} C_g p_g^G - \sum_{d=1}^{N_D} U_d p_d^D \quad (2a) \end{aligned}$$

subject to

$$(1b)-(1d) \quad (2b)$$

It is a simple linear program!

- One recognize a so-called **Linear Program (LP)**, here in a compact form):

$$\begin{array}{ll} \text{Min} & \mathbf{c}^T \mathbf{x} \\ & \mathbf{x} \geq 0 \end{array} \quad (3a)$$

subject to:

$$\mathbf{Ax} = \mathbf{b} \quad (3b)$$

$$\mathbf{Cx} \leq \mathbf{d} \quad (3c)$$

$$\mathbf{x} \geq 0 \quad (3d)$$

- LP problems can be readily solved in
 - **Python** with Pyomo package (<https://www.pyomo.org/>) → **Gurobi, CPLEX, ...**
 - **Julia** with JuMP package (<https://jump.dev/>) → **Gurobi, CPLEX, ...**
 - **GAMS** (<https://www.gams.com/>)
 - **Matlab**, for instance with the function linprog,
 - **R**, with the library/function lp_solve,

Getting the complete market-clearing

- By complete market-clearing is meant obtaining
 - **The schedule** for all supply and demand offers, as well as
 - **The price** at which the market is cleared, i.e., the so-called market-clearing or system price (in, e.g., Nord Pool)
- Only one **equality** constraint, i.e.,

$$\sum_d p_d^D - \sum_g p_g^G = 0$$

for which the associated Lagrange λ^S represents the system price.

- And $N_G + N_D$ **inequality** constraints:

$$0 \leq p_g^G \leq \bar{p}_g^G, \quad g = 1 \dots N_G; \quad 0 \leq p_d^D \leq \bar{p}_d^D, \quad d = 1 \dots N_D$$

for which the associated Lagrange multipliers γ_d^D and γ_g^G represents the unitary net benefits (i.e., surplus) for the various consumer and producer offer if the market is cleared at λ^S .

[NB: Most optimization functions and tools readily give you the solution of dual problems when solving the primal ones! E.g., see documentation of Pyomo ([Link](#))]

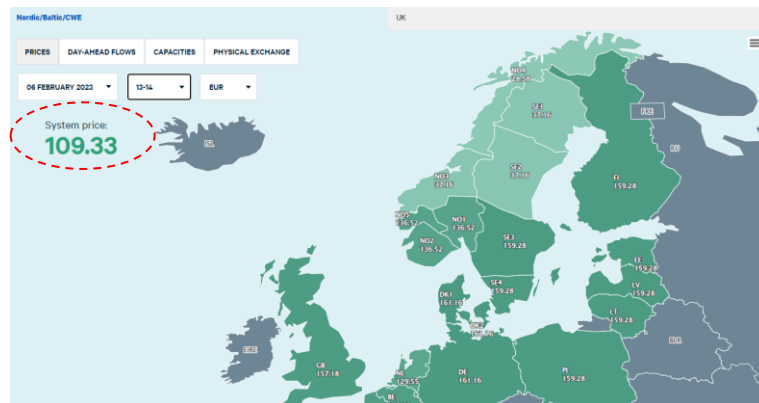
Optimization model for MC

- **Question:**

How to obtain market-clearing price within the optimization problem?

- **Answer:**

- ✓ The **dual variable** of the power balance equality, λ^S , provides the market-clearing price (“uniform” pricing scheme)!
- ✓ It represents the marginal cost for an adjustment of the Power balance equality constraint



Our example auction setup

Supply: (for a total of 1435 MWh)

Company	Supply/Demand	id	P_g^G (MW)	C_g^G (€/MWh)
C1	Supply	G1	120	0
C2	Supply	G2	50	0
C3	Supply	G3	200	15
C1	Supply	G4	400	30
C4	Supply	G5	60	32.5
C4	Supply	G6	50	34
C4	Supply	G7	60	36
C5	Supply	G8	100	37.5
C5	Supply	G9	70	39
C5	Supply	G10	50	40
C1	Supply	G11	70	60
C1	Supply	G12	45	70
C6	Supply	G13	50	100
C6	Supply	G14	60	150
C6	Supply	G15	50	200

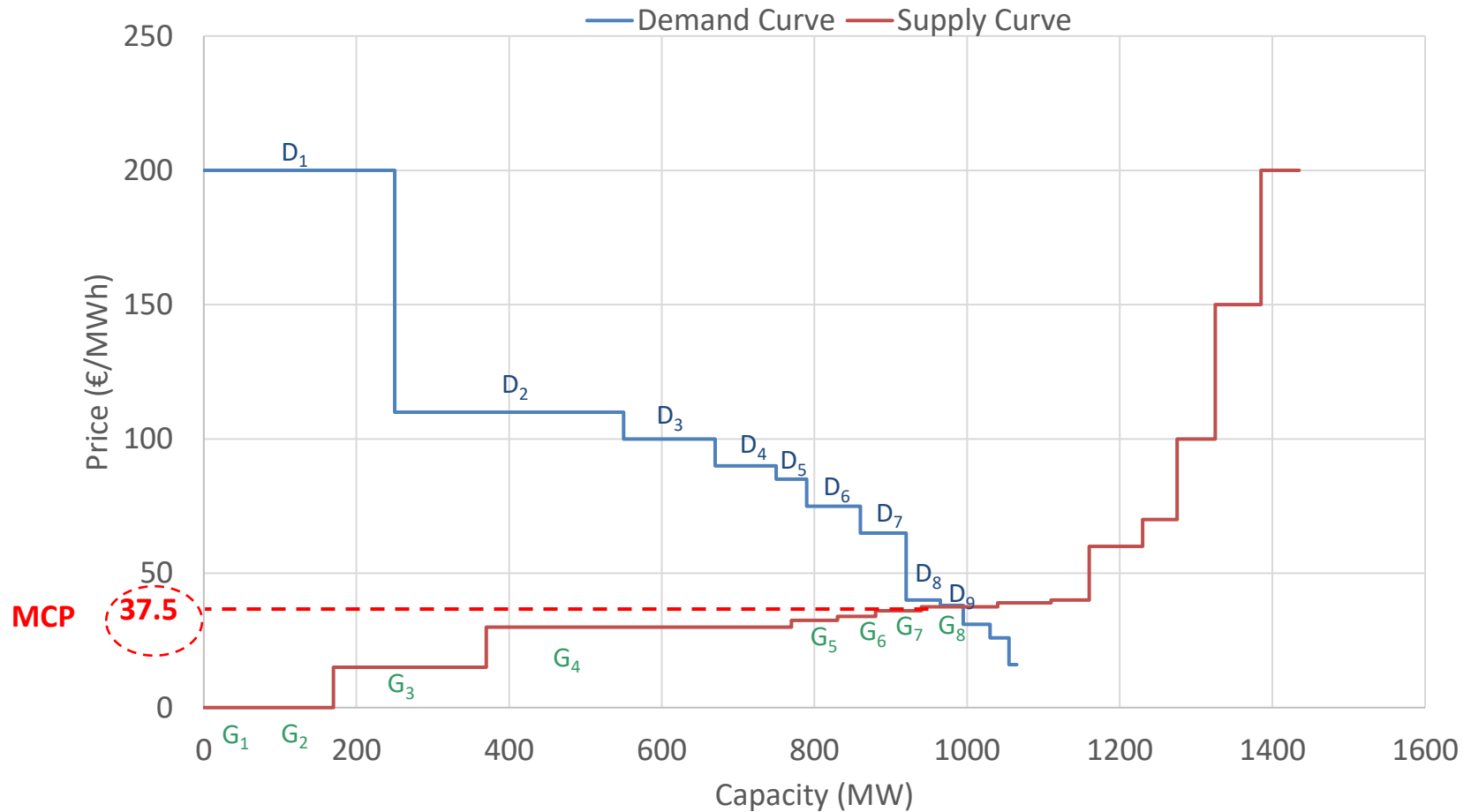
Our example auction setup

Demand: (for a total of 1065 MWh)

Company	Supply/Demand	id	P_d^D (MW)	U_d^D (€/MWh)
R1	Demand	D1	250	200
R2	Demand	D2	300	110
R3	Demand	D3	120	100
R4	Demand	D4	80	90
R5	Demand	D5	40	85
R6	Demand	D6	70	75
R1	Demand	D7	60	65
R5	Demand	D8	45	40
R4	Demand	D9	30	38
R5	Demand	D10	35	31
R1	Demand	D11	25	26
R2	Demand	D12	10	16

That is many offers to match... Could an optimisation problem readily give us the solution?

Supply and Demand Curve



Results from market clearing

The schedule for all supply and demand

```

===== Optimal Solution =====
Social Welfare (SW): 94817.5
Generation by G1: 120.0 MW
Generation by G2: 50.0 MW
Generation by G3: 200.0 MW
Generation by G4: 400.0 MW
Generation by G5: 60.0 MW
Generation by G6: 50.0 MW
Generation by G7: 60.0 MW
Generation by G8: 55.0 MW
Generation by G9: 0.0 MW
Generation by G10: 0.0 MW
Generation by G11: 0.0 MW
Generation by G12: 0.0 MW
Generation by G13: 0.0 MW
Generation by G14: 0.0 MW
Generation by G15: 0.0 MW
Demand by D1: 250.0 MW
Demand by D2: 300.0 MW
Demand by D3: 120.0 MW
Demand by D4: 80.0 MW
Demand by D5: 40.0 MW
Demand by D6: 70.0 MW
Demand by D7: 60.0 MW
Demand by D8: 45.0 MW
Demand by D9: 30.0 MW
Demand by D10: 0.0 MW
Demand by D11: 0.0 MW
Demand by D12: 0.0 MW
Market Clearing Price of Electricity: 37.5
    
```

unitary producers' surplus

unitary consumers' surplus

```

Dual values for generator capacity constraints:
Generator G1: 37.5
Generator G2: 37.5
Generator G3: 22.5
Generator G4: 7.5
Generator G5: 5.0
Generator G6: 3.5
Generator G7: 1.5
Generator G8: 0.0
Generator G9: 0.0
Generator G10: 0.0
Generator G11: 0.0
Generator G12: 0.0
Generator G13: 0.0
Generator G14: 0.0
Generator G15: 0.0
Dual values for demand capacity constraints:
Demand D1: 162.5
Demand D2: 72.5
Demand D3: 62.5
Demand D4: 52.5
Demand D5: 47.5
Demand D6: 37.5
Demand D7: 27.5
Demand D8: 2.5
Demand D9: 0.5
Demand D10: 0.0
Demand D11: 0.0
Demand D12: 0.0
    
```

Dual value for power balance equation

Producer Surplus for G1 $\rightarrow 120 \times 37.5 = 4500$ \$

Results-Elastic Demand

- Solving the primal LP for obtaining the supply and demand schedules yields:

Supply id.	Schedule (MWh)		Demand id.	Schedule (MWh)
G1	120		D1	250
G2	50		D2	300
G3	200		D3	120
G4	400		D4	80
G5	60		D5	40
G6	50		D6	70
G7	60		D7	60
G8	55		D8	45
G9-G15	0		D9	30
			D10-D12	0

for a total amount of energy scheduled of 995 MWh

- Solving the dual LP gives a system price of 37.5 €/MWh which corresponds to the price offer of G_8

Producer and Consumer Surplus

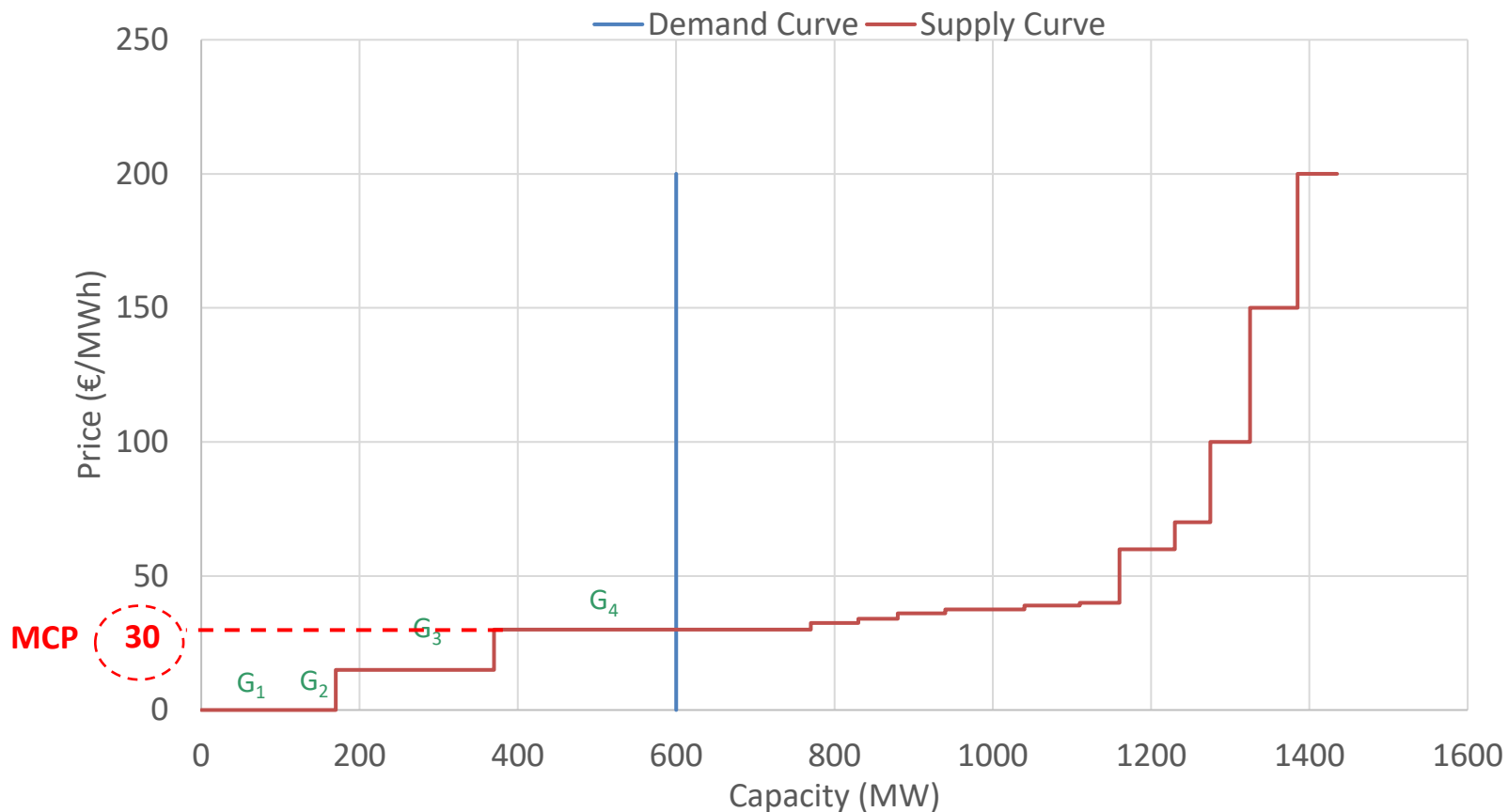
Generator	Dispatch (MW)	Unitary surplus (\$/MW)	Producer Surplus (\$)
G1	120	37.5	4500
G2	50	37.5	1875
G3	200	22.5	4500
G4	400	7.5	3000
G5	60	5	300
G6	50	3.5	175
G7	60	1.5	90
G8	55	0	0
G9	0	0	0
G10	0	0	0
G11	0	0	0
G12	0	0	0
G13	0	0	0
G14	0	0	0
G15	0	0	0
Total Producer Surplus (\$)			14440

Demand	Dispatch (MW)	Unitary surplus (\$/MW)	Consumer Surplus (\$)
D1	250	162.5	40625
D2	300	72.5	21750
D3	120	62.5	7500
D4	80	52.5	4200
D5	40	47.5	1900
D6	70	37.5	2625
D7	60	27.5	1650
D8	45	2.5	112.5
D9	30	0.5	15
D10	0	0	0
D11	0	0	0
D12	0	0	0
Total Consumer Surplus (\$)			80377.5

PS+CS= 94817.5 \$, which is equal to the objective value, social welfare

Inelastic Demand

- We fixed the demand at 600 MW



Inelastic Demand-Results from market clearing

The schedule for all supply and demand

```
===== Optimal Solution =====  
Social Welfare (SW): 110100.0  
Generation by G1: 120.0 MW  
Generation by G2: 50.0 MW  
Generation by G3: 200.0 MW  
Generation by G4: 230.0 MW  
Generation by G5: 0.0 MW  
Generation by G6: 0.0 MW  
Generation by G7: 0.0 MW  
Generation by G8: 0.0 MW  
Generation by G9: 0.0 MW  
Generation by G10: 0.0 MW  
Generation by G11: 0.0 MW  
Generation by G12: 0.0 MW  
Generation by G13: 0.0 MW  
Generation by G14: 0.0 MW  
Generation by G15: 0.0 MW  
Demand by D1: 600.0 MW  
Market Clearing Price of Electricity: 30.0
```

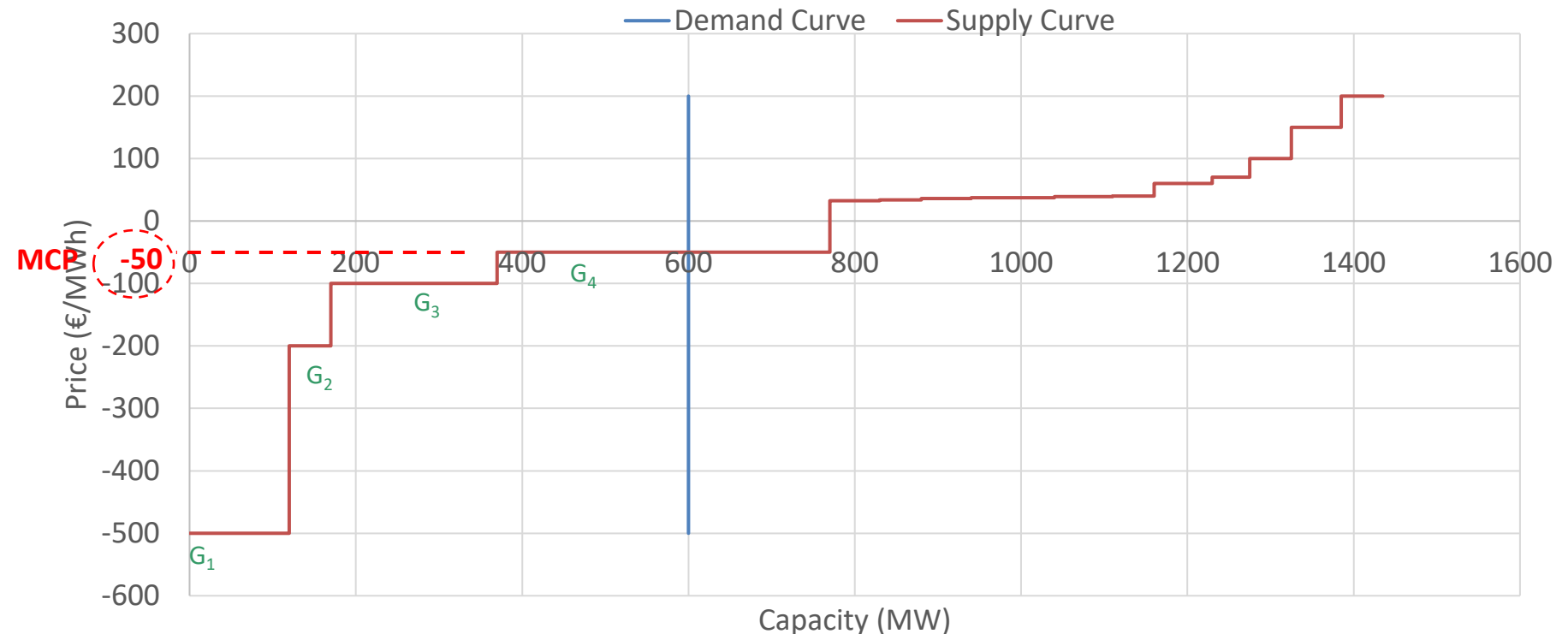
Dual value for power balance equation

unitary generators's

```
Dual values for generator capacity constraints:  
Generator G1: 30.0  
Generator G2: 30.0  
Generator G3: 15.0  
Generator G4: 0.0  
Generator G5: 0.0  
Generator G6: 0.0  
Generator G7: 0.0  
Generator G8: 0.0  
Generator G9: 0.0  
Generator G10: 0.0  
Generator G11: 0.0  
Generator G12: 0.0  
Generator G13: 0.0  
Generator G14: 0.0  
Generator G15: 0.0
```

Negative prices

- Nordpool: Single hourly orders need to be defined with at least two price steps, minimum price (EUR -500) and maximum price (EUR 4000).



<https://www.nordpoolgroup.com/en/trading/Day-ahead-trading/Order-types/Hourly-bid/>

Negative prices-Results from market clearing

The schedule for all supply and demand

```
===== Optimal Solution =====  
Social Welfare (SW): 221500.0  
Generation by G1: 120.0 MW  
Generation by G2: 50.0 MW  
Generation by G3: 200.0 MW  
Generation by G4: 230.0 MW  
Generation by G5: 0.0 MW  
Generation by G6: 0.0 MW  
Generation by G7: 0.0 MW  
Generation by G8: 0.0 MW  
Generation by G9: 0.0 MW  
Generation by G10: 0.0 MW  
Generation by G11: 0.0 MW  
Generation by G12: 0.0 MW  
Generation by G13: 0.0 MW  
Generation by G14: 0.0 MW  
Generation by G15: 0.0 MW  
Demand by D1: 600.0 MW  
Market Clearing Price of Electricity: -50.0
```

Dual value for power balance equation

unitary generators' surplus

```
Dual values for generator capacity constraints:  
Generator G1: 450.0  
Generator G2: 150.0  
Generator G3: 50.0  
Generator G4: 0.0  
Generator G5: 0.0  
Generator G6: 0.0  
Generator G7: 0.0  
Generator G8: 0.0  
Generator G9: 0.0  
Generator G10: 0.0  
Generator G11: 0.0  
Generator G12: 0.0  
Generator G13: 0.0  
Generator G14: 0.0  
Generator G15: 0.0
```

Write the dual problem

$$\mathbf{max}_{x_1 \geq 0, x_2 \leq 0, x_3} (v_1 x_1 + v_2 x_2 + v_3 x_3)$$

$$\text{subject to } a_1 x_1 + x_2 + x_3 \leq b_1$$

$$x_1 + a_2 x_2 = b_2$$

$$a_3 x_3 \geq b_3$$



Primal problem

- **Step 1.** If necessary, rewrite the objective as a minimization.
- **Step 2.** Rewrite each inequality constraint as a "less than or equal", and rearrange each constraint so that the right-hand side is 0.

$$\mathbf{min}_{x_1 \geq 0, x_2 \leq 0, x_3} (-v_1 x_1 - v_2 x_2 - v_3 x_3)$$

$$\text{subject to } a_1 x_1 + x_2 + x_3 - b_1 \leq 0$$

$$x_1 + a_2 x_2 - b_2 = 0$$

$$-a_3 x_3 + b_3 \leq 0$$

[To retrieve the dual LP, follow: Lahaie S (2015). How to take the dual of a Linear Program. ([link](#))]

Write the dual problem

- **Step 3.** Define a non-negative dual variable for each inequality constraint, and an unrestricted dual variable for each equality constraint.

$$\begin{array}{ll}
 \min_{x_1 \geq 0, x_2 \leq 0, x_3} & (-v_1 x_1 - v_2 x_2 - v_3 x_3) \\
 \text{subject to:} & a_1 x_1 + x_2 + x_3 - b_1 \leq 0 \quad (\lambda_1) \\
 & x_1 + a_2 x_2 - b_2 = 0 \quad (\lambda_2) \\
 & -a_3 x_3 + b_3 \leq 0 \quad (\lambda_3)
 \end{array}
 \quad \longrightarrow \quad
 \begin{array}{l}
 \lambda_1 \geq 0, \lambda_2 \text{ unrestricted}, \lambda_3 \geq 0
 \end{array}$$

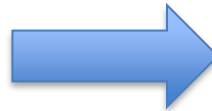
- **Step 4.** For each constraint, eliminate it and add the term (dual variable)(left-hand side of constraint) to the objective. Maximize the result over the dual variables.

$$\begin{array}{l}
 \max_{\lambda_1 \geq 0, \lambda_2, \lambda_3 \geq 0} \min_{x_1 \geq 0, x_2 \leq 0, x_3} (-v_1 x_1 - v_2 x_2 - v_3 x_3) \\
 \quad + \lambda_1 (a_1 x_1 + x_2 + x_3 - b_1) \\
 \quad + \lambda_2 (x_1 + a_2 x_2 - b_2) \\
 \quad + \lambda_3 (-a_3 x_3 + b_3)
 \end{array}$$

Write the dual problem

- **Step 5.** Rewrite the objective so that it consists of several terms of the form (primal variable)(expression with dual variables), plus remaining terms involving only dual variables.

$$\begin{aligned} \max_{\lambda_1 \geq 0, \lambda_2, \lambda_3 \geq 0} \min_{x_1 \geq 0, x_2 \leq 0, x_3} & (-v_1 x_1 - v_2 x_2 - v_3 x_3) \\ & + \lambda_1 (a_1 x_1 + x_2 + x_3 - b_1) \\ & + \lambda_2 (x_1 + a_2 x_2 - b_2) \\ & + \lambda_3 (-a_3 x_3 + b_3) \end{aligned}$$



$$\begin{aligned} \max_{\lambda_1 \geq 0, \lambda_2, \lambda_3 \geq 0} \min_{x_1 \geq 0, x_2 \leq 0, x_3} & (-b_1 \lambda_1 - b_2 \lambda_2 + b_3 \lambda_3) \\ & + x_1 (a_1 \lambda_1 + \lambda_2 - v_1) \\ & + x_2 (\lambda_1 + a_2 \lambda_2 - v_2) \\ & + x_3 (\lambda_1 - a_3 \lambda_3 - v_3) \end{aligned}$$

Write the dual problem

- **Step 6.** Remove each term of the form (primal variable)(expression with dual variables) and replace it with a constraint of the form:
 - expression ≥ 0 , if the primal variable is non-negative.
 - Expression ≤ 0 , if the primal variable is non-positive.
 - expression = 0, if the primal variable is unrestricted.

$$\max_{\lambda_1 \geq 0, \lambda_2, \lambda_3 \geq 0} \min_{x_1 \geq 0, x_2 \leq 0, x_3} (-b_1 \lambda_1 - b_2 \lambda_2 + b_3 \lambda_3)$$

$$+x_1(a_1 \lambda_1 + \lambda_2 - v_1)$$

$$+x_2(\lambda_1 + a_2 \lambda_2 - v_2)$$

$$+x_3(\lambda_1 - a_3 \lambda_3 - v_3)$$



$$\max_{\lambda_1 \geq 0, \lambda_2, \lambda_3 \geq 0} (-b_1 \lambda_1 - b_2 \lambda_2 + b_3 \lambda_3)$$

Subject to:

$$a_1 \lambda_1 + \lambda_2 - v_1 \geq 0$$

$$\lambda_1 + a_2 \lambda_2 - v_2 \leq 0$$

$$\lambda_1 - a_3 \lambda_3 - v_3 = 0$$

Write the dual problem

- **Step 7.** If the original LP was a maximization rewritten as a minimization in Step 1, rewrite the result of the previous step as a minimization.

$$\mathbf{max}_{x_1 \geq 0, x_2 \leq 0, x_3} (v_1 x_1 + v_2 x_2 + v_3 x_3)$$

$$\begin{aligned} \text{subject to: } & a_1 x_1 + x_2 + x_3 \leq b_1 \\ & x_1 + a_2 x_2 = b_2 \\ & a_3 x_3 \geq b_3 \end{aligned}$$

Primal Problem

$$\mathbf{min}_{\lambda_1 \geq 0, \lambda_2 \leq 0, \lambda_3 \geq 0} (b_1 \lambda_1 + b_2 \lambda_2 - b_3 \lambda_3)$$

Subject to:

$$\begin{aligned} & a_1 \lambda_1 + \lambda_2 \geq v_1 \\ & \lambda_1 + a_2 \lambda_2 \leq v_2 \\ & \lambda_1 - a_3 \lambda_3 = v_3 \end{aligned}$$

Dual Problem

Write the dual of the market clearing LP

$$\min_{x \geq 0} (c^T x)$$

$$\text{s.t. } Ax = b \quad \lambda$$

$$Cx \leq d \quad \gamma$$

Primal Problem

S1, S2

$$\min_{x \geq 0} (c^T x)$$

$$\text{s.t. } Ax - b = 0 \quad \lambda$$

$$Cx - d \leq 0 \quad \gamma$$

S3, S4

$$\max_{\lambda, \gamma \geq 0} \min_{x \geq 0} (c^T x)$$

$$+ \lambda^T (Ax - b)$$

$$+ \gamma^T (Cx - d)$$

S5, S4

$$\max_{\lambda, \gamma \geq 0} \min_{x \geq 0} (-\lambda^T b - \gamma^T d)$$

$$+ (c + A^T \lambda + C^T \gamma)^T x$$

S6

$$\max_{\lambda, \gamma \geq 0} (-\lambda^T b - \gamma^T d)$$

$$\text{s.t. } c + A^T \lambda + C^T \gamma \geq 0$$

Dual Problem

More specifically for the market-clearing problem

$$\begin{array}{ccc}
 \text{Primal} & \left\{ \begin{array}{l} \mathbf{\min}_{x \geq 0} (c^T x) \\ \text{s.t. } Ax = b \quad \lambda \\ \quad Cx \leq d \quad \gamma \end{array} \right. & \xrightarrow{\text{Dual}} & \left\{ \begin{array}{l} \mathbf{\max}_{\lambda, \gamma \geq 0} (-\lambda^T b - \gamma^T d) \\ \text{s.t. } c + A^T \lambda + C^T \gamma \geq 0 \end{array} \right. \text{Dual}
 \end{array}$$

- The market clear problem is formulated as bellow:

$$\begin{array}{ccc}
 \text{Primal} & \left\{ \begin{array}{l} \text{Max}_{p_g^G \geq 0, p_d^D \geq 0} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G \\ \text{s.t.: } \sum_{d=1}^{N_D} p_d^D - \sum_{g=1}^{N_G} p_g^G = 0 \quad (\lambda) \\ \quad 0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \quad (\gamma_g^G) \\ \quad 0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \quad (\gamma_d^D) \end{array} \right. & \xrightarrow{\text{Standard Form}} & \left\{ \begin{array}{l} \text{Min}_{p_g^G \geq 0, p_d^D \geq 0} SW = \sum_{g=1}^{N_G} C_g p_g^G - \sum_{d=1}^{N_D} U_d p_d^D \\ \text{s.t.: } \sum_{d=1}^{N_D} p_d^D - \sum_{g=1}^{N_G} p_g^G = 0 \quad (\lambda) \\ \quad 0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G \quad (\gamma_g^G) \\ \quad 0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D \quad (\gamma_d^D) \end{array} \right.
 \end{array}$$

- The dual of the market clearing LP is also an LP which writes

$$\begin{array}{ccc}
 \begin{array}{l} \text{Max}_{\lambda^s, \gamma_g^G \geq 0, \gamma_d^D \geq 0} - \sum_g \gamma_g^G \bar{P}_g^G - \sum_d \gamma_d^D \bar{P}_d^D \\ \text{s.t.: } C_g - \lambda^s + \gamma_g^G \geq 0, \quad g = 1 \dots N_G \\ \quad -U_d + \lambda^s + \gamma_d^D \geq 0, \quad d = 1 \dots N_D \end{array} & \xrightarrow{\quad} & \left\{ \begin{array}{l} \text{Max}_{\lambda^s, \gamma_g^G \geq 0, \gamma_d^D \geq 0} - \sum_g \gamma_g^G \bar{P}_g^G - \sum_d \gamma_d^D \bar{P}_d^D \\ \text{s.t.: } \lambda^s - \gamma_g^G \leq C_g, \quad g = 1 \dots N_G \\ \quad -\lambda^s - \gamma_d^D \leq -U_d, \quad d = 1 \dots N_D \end{array} \right. \text{Dual}
 \end{array}$$

[To retrieve the dual LP, follow: Lahaie S (2015). How to take the dual of a Linear Program. ([link](#))]

Results from dual market clearing

System price

unitary producers' surplus

unitary consumers' surplus

```
===== Optimal Solution of Dual Market Clearing =====
Social Welfare (SW): .94817 5
System Price (lambda): 37.5
Unitary Producer Surplus for G1: 37.5 MW
Unitary Producer Surplus for G2: 37.5 MW
Unitary Producer Surplus for G3: 22.5 MW
Unitary Producer Surplus for G4: 7.5 MW
Unitary Producer Surplus for G5: 5.0 MW
Unitary Producer Surplus for G6: 3.5 MW
Unitary Producer Surplus for G7: 1.5 MW
Unitary Producer Surplus for G8: 0.0 MW
Unitary Producer Surplus for G9: 0.0 MW
Unitary Producer Surplus for G10: 0.0 MW
Unitary Producer Surplus for G11: 0.0 MW
Unitary Producer Surplus for G12: 0.0 MW
Unitary Producer Surplus for G13: 0.0 MW
Unitary Producer Surplus for G14: 0.0 MW
Unitary Producer Surplus for G15: 0.0 MW
Unitary Consumers Surplus for D1: 162.5 MW
Unitary Consumers Surplus for D2: 72.5 MW
Unitary Consumers Surplus for D3: 62.5 MW
Unitary Consumers Surplus for D4: 52.5 MW
Unitary Consumers Surplus for D5: 47.5 MW
Unitary Consumers Surplus for D6: 37.5 MW
Unitary Consumers Surplus for D7: 27.5 MW
Unitary Consumers Surplus for D8: 2.5 MW
Unitary Consumers Surplus for D9: 0.5 MW
Unitary Consumers Surplus for D10: 0.0 MW
Unitary Consumers Surplus for D11: 0.0 MW
Unitary Consumers Surplus for D12: 0.0 MW
```

The schedule for Supply

The schedule for demand

```
Dual values for generator capacity constraints:
Generator G1: 120.0
Generator G2: 50.0
Generator G3: 200.0
Generator G4: 400.0
Generator G5: 60.0
Generator G6: 50.0
Generator G7: 60.0
Generator G8: 55.0
Generator G9: 0.0
Generator G10: 0.0
Generator G11: 0.0
Generator G12: 0.0
Generator G13: 0.0
Generator G14: 0.0
Generator G15: 0.0
Dual values for demand capacity constraints:
Demand D1: 250.0
Demand D2: 300.0
Demand D3: 120.0
Demand D4: 80.0
Demand D5: 40.0
Demand D6: 70.0
Demand D7: 60.0
Demand D8: 45.0
Demand D9: 30.0
Demand D10: 0.0
Demand D11: 0.0
Demand D12: 0.0
```

Energy only and capacity Markets

Energy-Only vs. Capacity Markets

- **Energy-only markets:** Generators get paid for the energy they produce; rely on price signals (often with potential for high scarcity prices) to encourage investment and secure reliability.
- **Capacity markets:** Generators get additional payments for available capacity; add an extra revenue stream through capacity payments, aiming to ensure adequate capacity and reduce revenue uncertainty for generators.

Energy-Only vs. Capacity Markets

Aspect	Energy-Only Market	Capacity Market
Revenue Streams	Generators rely entirely on energy prices (and ancillary services) for income	Generators earn capacity payments in addition to any revenue from energy sales
Key Motivation	Encourage investment through high prices during scarcity	Provide a steady payment to ensure enough capacity is built/retained for future needs
Price Behavior	Can experience price spikes during peak demand	More stable pricing for generators, as capacity payments offset low or volatile energy prices
Risk and Incentive	High risk for investors; they depend on fewer hours of high prices	Lower risk for investors; the capacity auction guarantees baseline revenue
System Reliability	Relies on scarcity pricing to incentivize new builds or peak reserves	Establishes firm commitments (obligations) for availability; penalizes non-performance
Examples	Norway, Germany, ERCOT (Texas)	France, PJM, UK

Determining the optimal level of capacity in capacity markets remains a complex task

MD- Small example - Energy Only vs Capacity Markets

Investment cost of gas turbine:	650 €/kW
For simplicity: assume variable cost	=0
Discount rate:	10 %
Required payback time:	15 yrs
From this: annuity (ϵ)	= 0,1314
Expected running time:	25 hours

What is the required price during those hours to make this a profitable investment?

MD- Small example- Solution

- Annual cost: $650000 \cdot 0.1314 = 85458 \text{ €/MW}$
- Required price: $85458 / 25 = 3418 \text{ €/MWh}$
- Not profitable with price cap of 3000 €/MWh (old price cap)!
- The new price caps are -500 €/MWh and 4000 €/MWh
- Much higher price cap is not politically acceptable

Characterization of Capacity Mechanisms

- What is the product
 - Physical capacity
 - Financial instrument
- How is the required amount determined
 - **Price-Based Determination:** Market finds the volume given an administratively set price curve¹.
 - **Top-Down Quantity:** Regulator or system operator pre-defines a precise capacity requirement.
 - **Bottom-Up with Central Check:** Each entity procures capacity to cover its own load; the central authority verifies and corrects.
 - **Individual Customer Choice:** Consumers buy the level of reliability/capacity they want, though system oversight remains necessary.

¹The authority publishes a “capacity demand curve”—for instance, the “Variable Resource Requirement (VRR)” in some U.S. capacity markets—indicating how much capacity the system wants to buy at different prices.

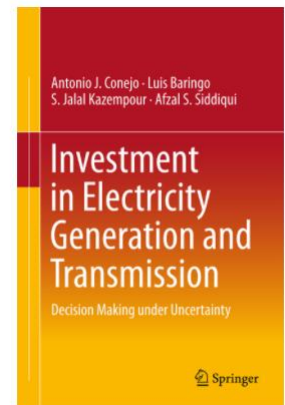
Screening model for optimal generation mix

Reading material:

- Ivar Wangensteen, Power System Economics: The Nordic Electricity Market
 - Chapter 6.7, Optimal mix of generating units

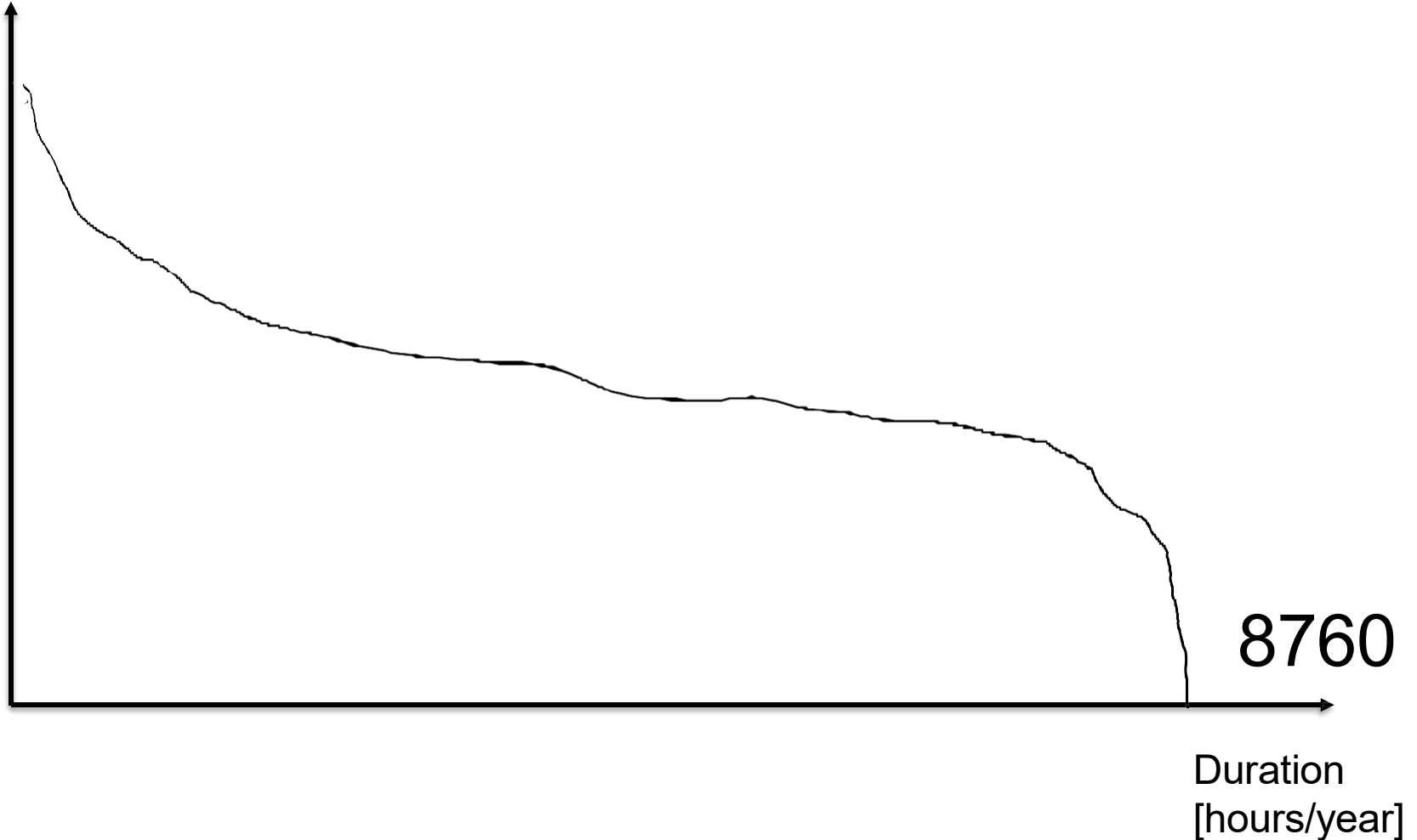
The screening model

- Determines optimal mix of generating unit under equilibrium condition
- Simplified investment planning model from the social planner's perspective
 - Uncertainty not included
 - Time-coupling constraints not included
- However: realistic numerical models are large and complicated and must also use approximations
- The screening model gives insight into the principle factors that determine ratios of installed capacities



The load duration curve

Load [GWh/h]



Assume two generation assets

- Base load unit:
 - High investment cost
 - Low marginal generation cost
 - Example: Nuclear power plant
- Peak load unit:
 - Comparatively low investment cost
 - High marginal generation cost
 - Example: Gas open cycle power plant

Cost minimisation problem

- The objective is to minimize total cost, given that the installed capacity must be able to cover load at all times.

$$\min C^{Tot} = C_B + C_P$$

$$C_B = k_B P_B + c_B W_B$$

$$C_P = k_P P_P + c_P W_P$$

Where

C_i - Cost (total) of technology i [NOK]

k_i - fixed cost of technology i [NOK/kW/year]

P_i - installed capacity of technology i [kW]

c_i - short run marginal cost of technology i [NOK/kWh]

W_i - produced electricity by technology i [kWh]

Equilibrium condition

- Of the combinations that cover the load, it can be shown that the optimal capacity split between the peak and baseload unit is given by

$$T = \frac{k_B - k_P}{c_P - c_B}$$

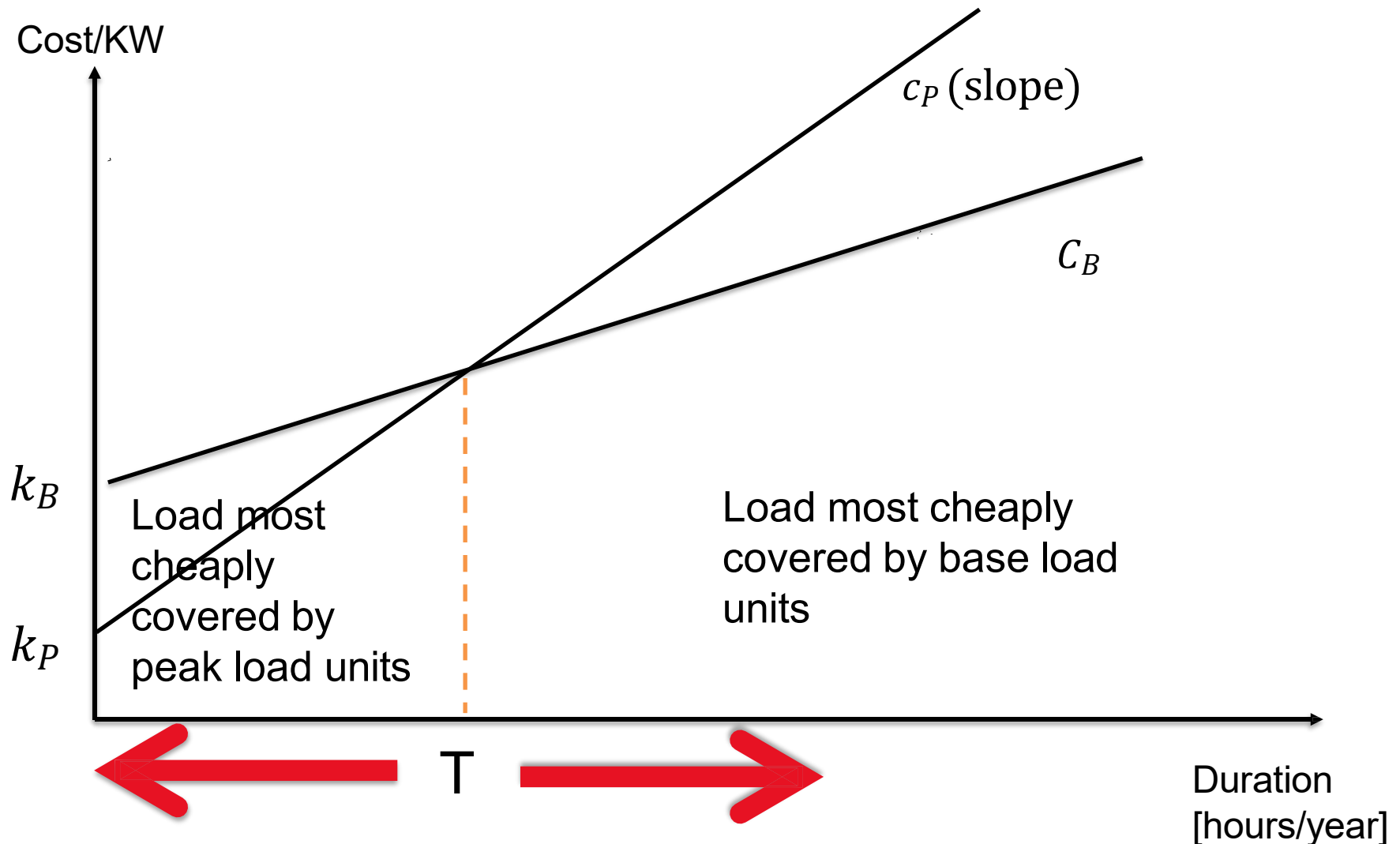
Where

T – time where peak load unit is needed to cover the load

See chapter 6.7.1 in Wangesteen 2011 for details.

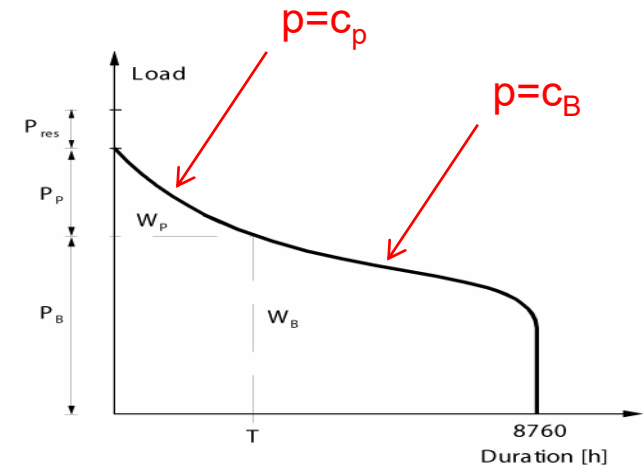
Intuition on next slide. 😊

Finding the optimal generation mix



Profits of generating units

- Profit = Income – Costs
- Income definition:



$$I = \left(\underbrace{W_p}_{\substack{\text{KWh} \\ \text{produced} \\ \text{by peak unit}}} + \underbrace{P_B T}_{\substack{\text{Base load} \\ \text{running} \\ \text{in peak hours}}} \right) c_P + \left(\underbrace{W_B}_{\substack{\text{KWh} \\ \text{produced} \\ \text{by base} \\ \text{unit}}} - P_B T \right) c_B$$

- Rearrange:

$$I = c_p W_p + c_p P_B T + c_B W_B - c_B P_B T$$

$$I = c_p W_p + c_B W_B + P_B T (c_p - c_B)$$

Profits of generating units

- We have established: $I = c_P W_P + c_B W_B + P_B T (c_P - c_B)$
- Remember: $T = \frac{k_B - k_P}{c_P - c_B}$
- This gives: $I = W_P c_P + W_B c_B + P_B (k_B - k_P)$

covers variable costs
does not cover fixed costs
- Fixed costs: $k_B P_B + k_P P_P$
- Missing revenue: $k_P (P_P + P_B) = (k_B P_B + k_P P_P) - (k_B P_B - k_P P_B)$
- The "missing money" problem

Revenue deficit per technology

- Average profit deficit for peak load units:

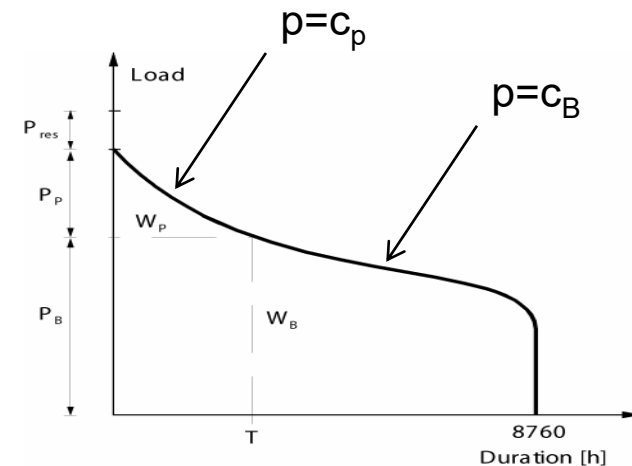
$$\frac{D_P}{P_P} = \frac{\overbrace{W_P c_p}^{\text{Revenue}} - \overbrace{(W_P c_p + P_P k_P)}^{\text{Cost}}}{P_P} = -k_P$$

- Average profit deficit for base load units:

$$\frac{D_B}{P_B} = \frac{\overbrace{(W_B - P_B T) c_B + P_B T c_P}^{\text{Revenue}} - \overbrace{(W_B c_B + P_B k_B)}^{\text{Cost}}}{P_B} = (c_P - c_B)T - k_B =$$

$$(c_P - c_B) \frac{k_B - k_P}{c_P - c_B} - k_B = -k_P$$

- Revenue loss per kW is the same for both technologies
 - In fact, the peaker does not make any profit
- Equal to the capacity cost of the peaker



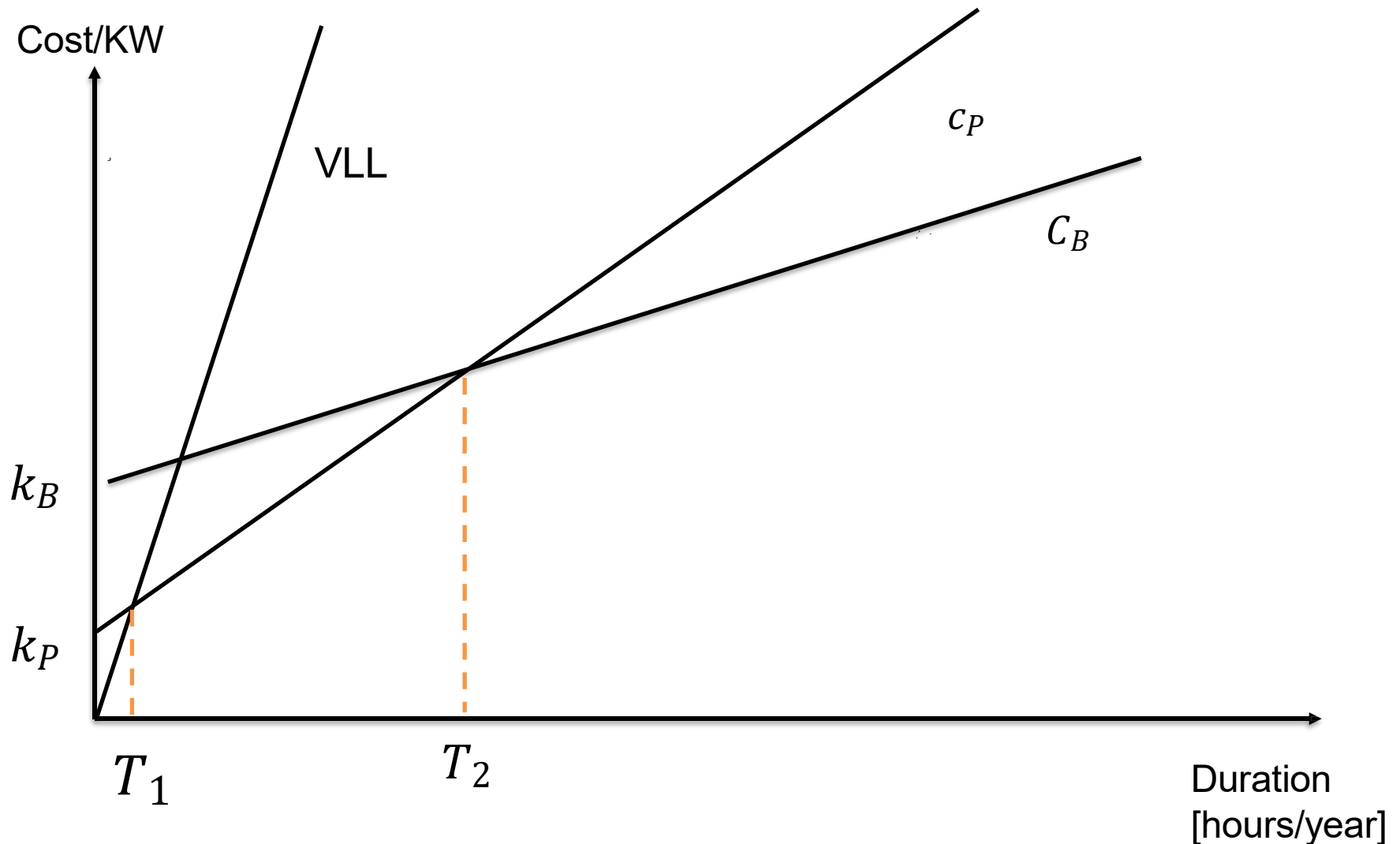
Missing money - intuition

- Competitive market gives price = marginal cost
- The peak unit only covers its variable cost, no contribution to cover fixed cost
- The base load unit gets some contribution to cover its fixed cost – but not sufficient.
- All generation units operate at a long-term loss

Solving the missing money problem: Introducing load shedding

- Introduce controlled load shedding as option in peak hours – corresponds to some price elasticity
- Load shedding has a very high variable cost (Value of Lost Load – VLL), but no investment costs

Optimal generation/shedding mix



Load shedding

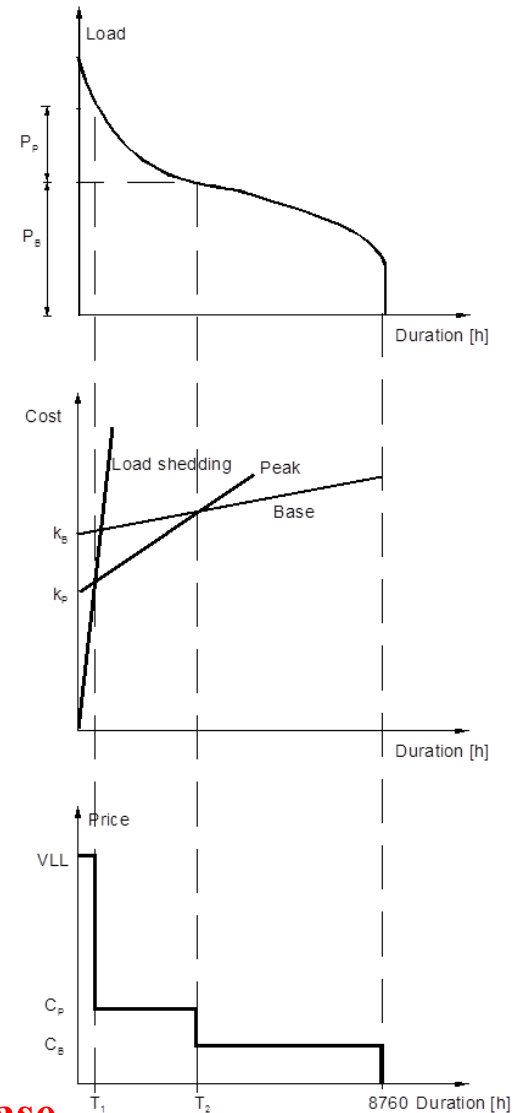
- Load shedding can mathematically be viewed as a third type of unit
- Sets the price during some peak hours
- Has zero capacity cost

$$D_P = (VLL - c_P) P_P T_1 = P_P k_P$$

$$\begin{aligned} D_B &= (VLL - c_B) P_B T_1 + (c_P - c_B) P_B (T_2 - T_1) = \\ &= [(VLL - c_P) T_1 + (c_P - c_B) T_2] P_B = \\ &= \left[\frac{(VLL - c_P) k_P}{VLL - c_P} + \frac{(c_P - c_B)(k_B - k_P)}{c_P - c_B} \right] P_B = k_B P_B \end{aligned}$$

The operation surplus covers the capital costs exactly in this case

$$T_1 = \frac{k_P}{VLL - c_P}, \quad T_2 = \frac{k_B - k_P}{c_P - c_B}$$



Summary – screening model

- With load shedding as option, both the peak load unit and base load unit covers its costs
- Load shedding does not equate black-out – it can be thought of as a (slight) price elasticity
- Energy only-market is sufficient, but:
 - Load shed must be an option
 - The public must accept high price spikes

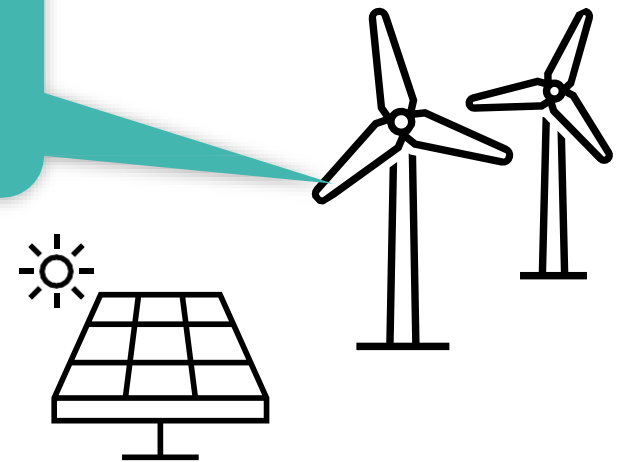


Screening model in a more renewable power system

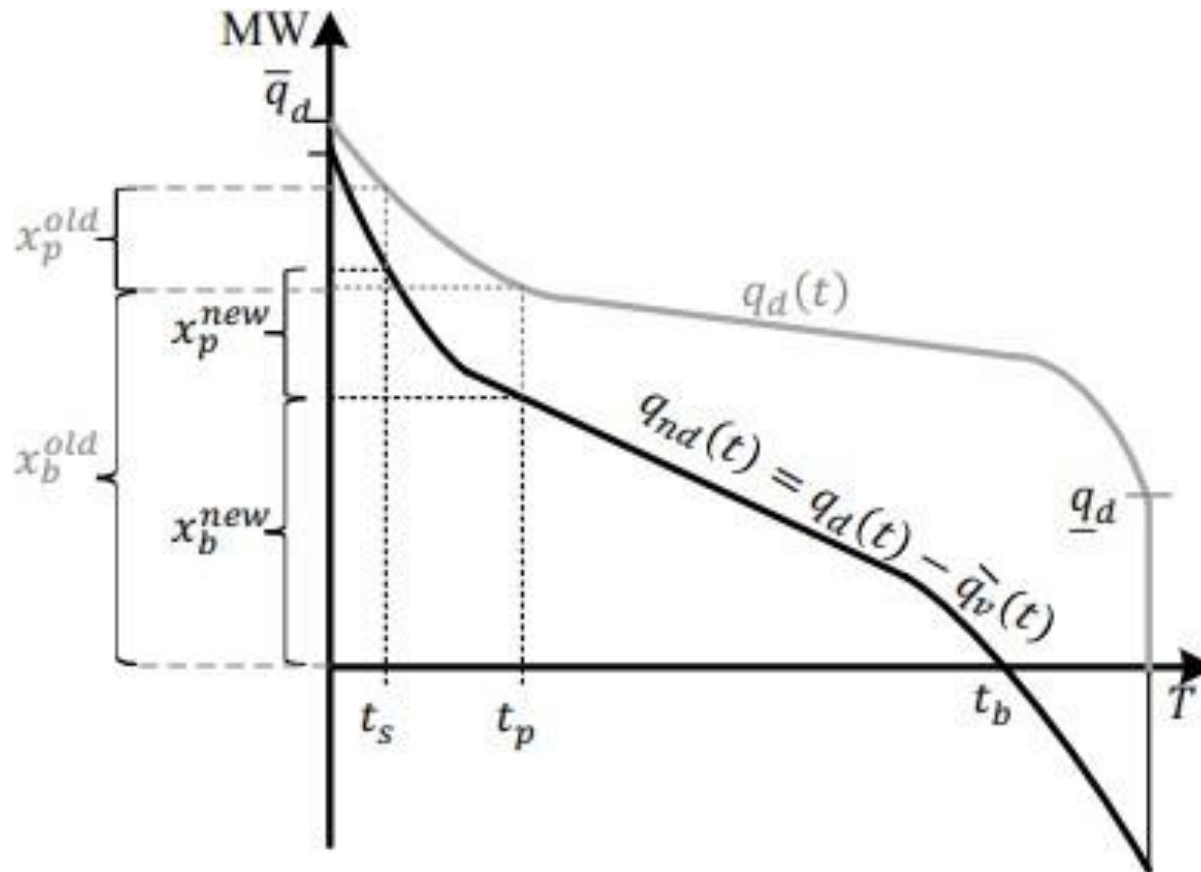
- With intermittent renewable energy sources we focus on the residual demand curve rather than the pure load duration curve
- Residual demand = load – wind – pv – run of river



Talking about base- and peak loaders are a bit old-fashioned, isn't it?



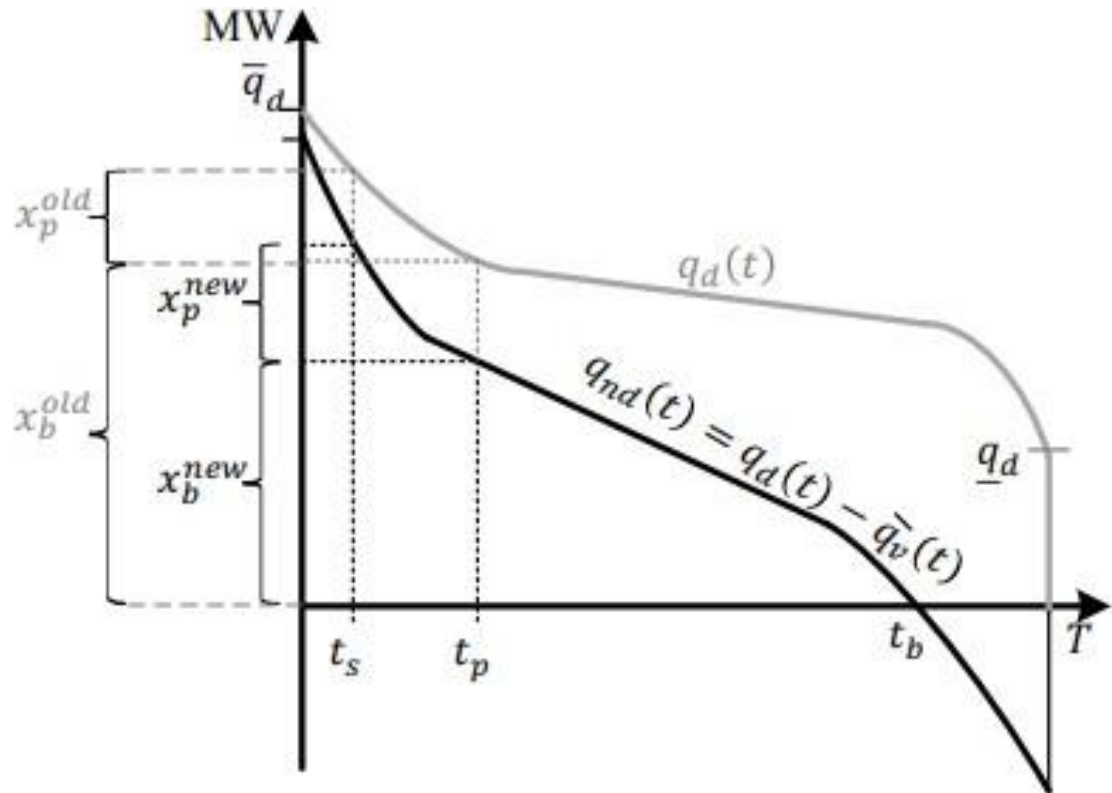
Residual demand curve - illustration



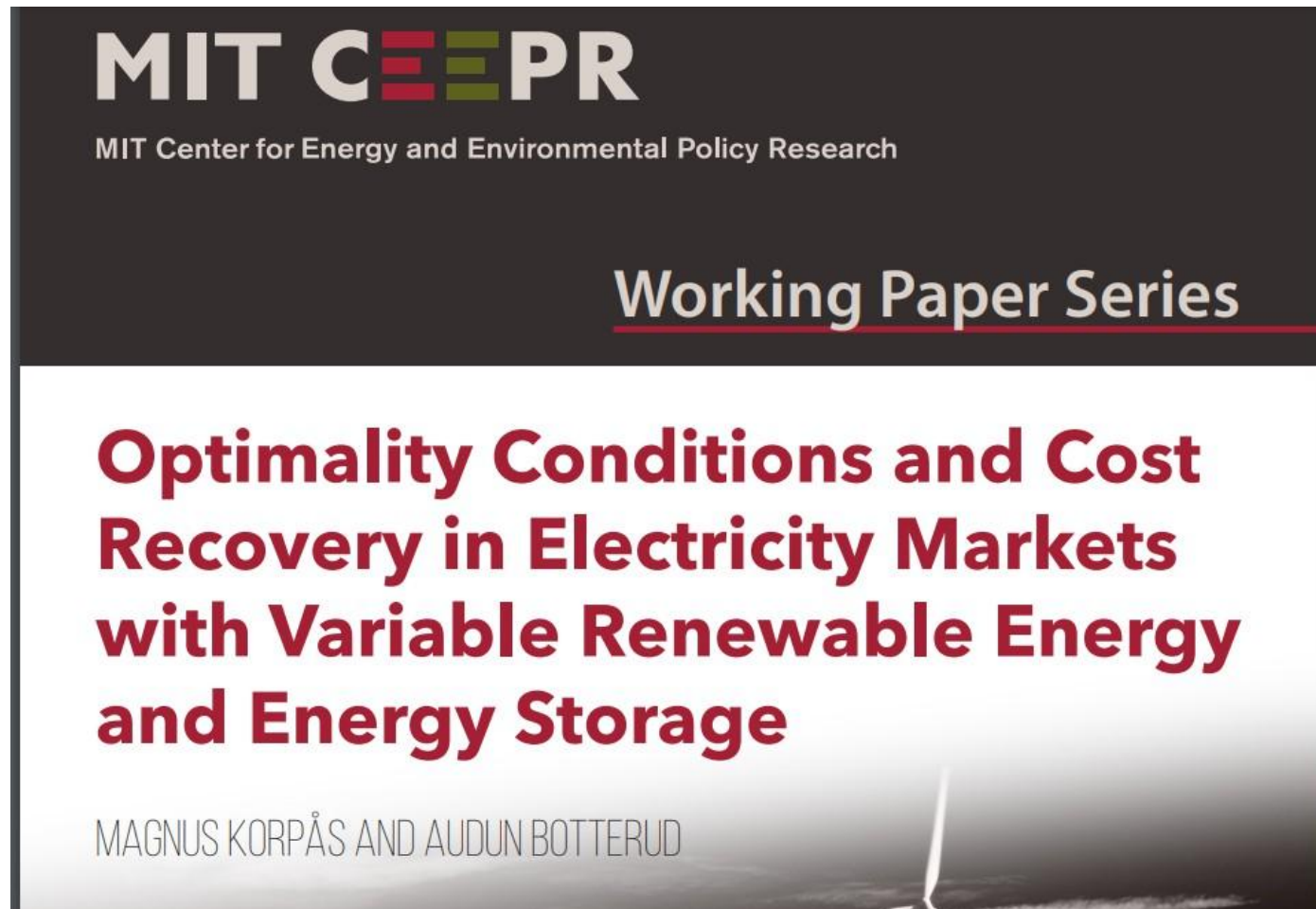
From Korpås and Botterud (2020)

Insights

- When residual load is 0 or negative, there will be zero-prices
- Presence of intermittent production will alter the optimal mix of conventional peaker and baseload units
- Korpås and Botterud argue that an energy-only market is still sufficient even with high penetration of renewables

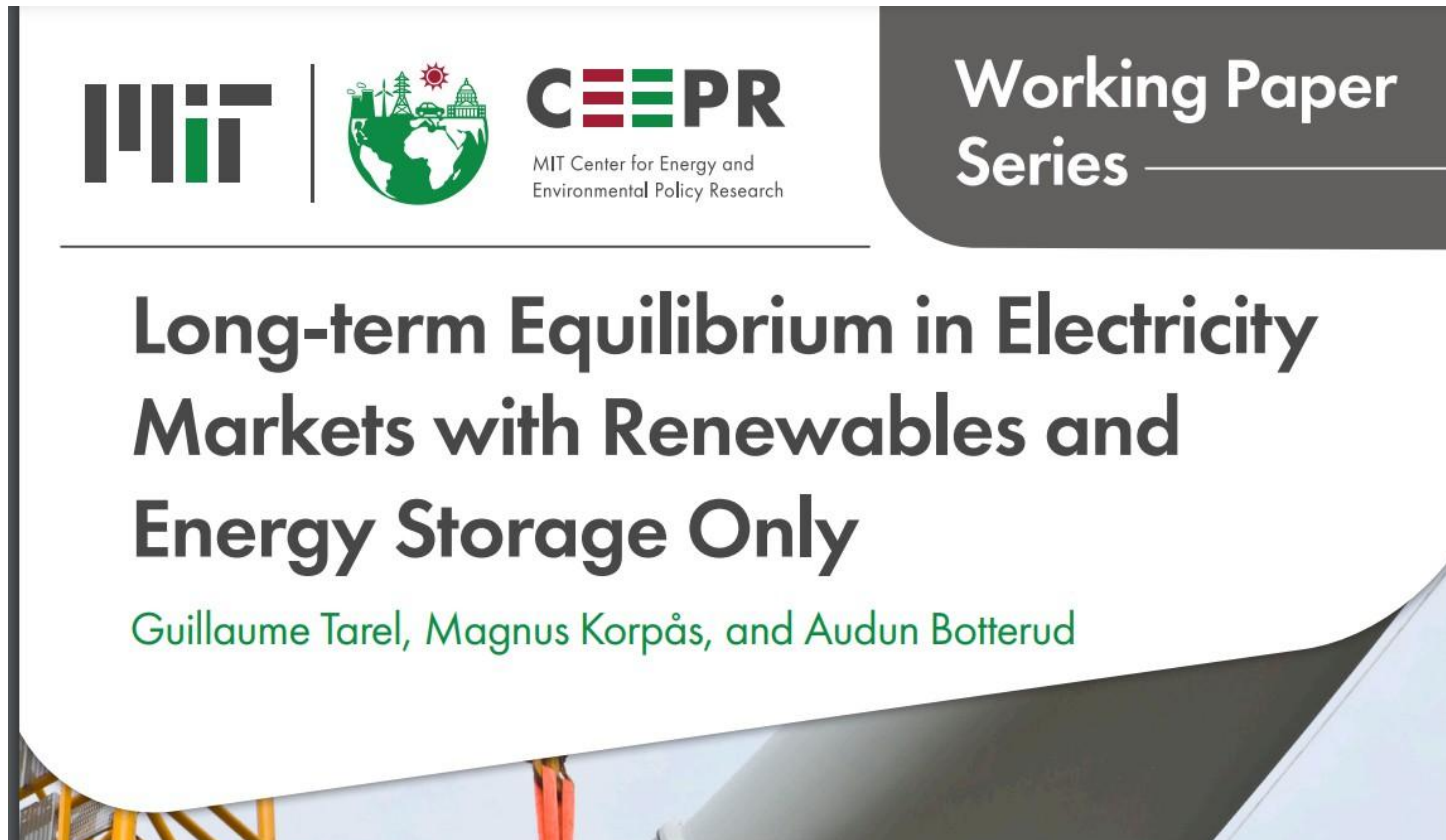


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